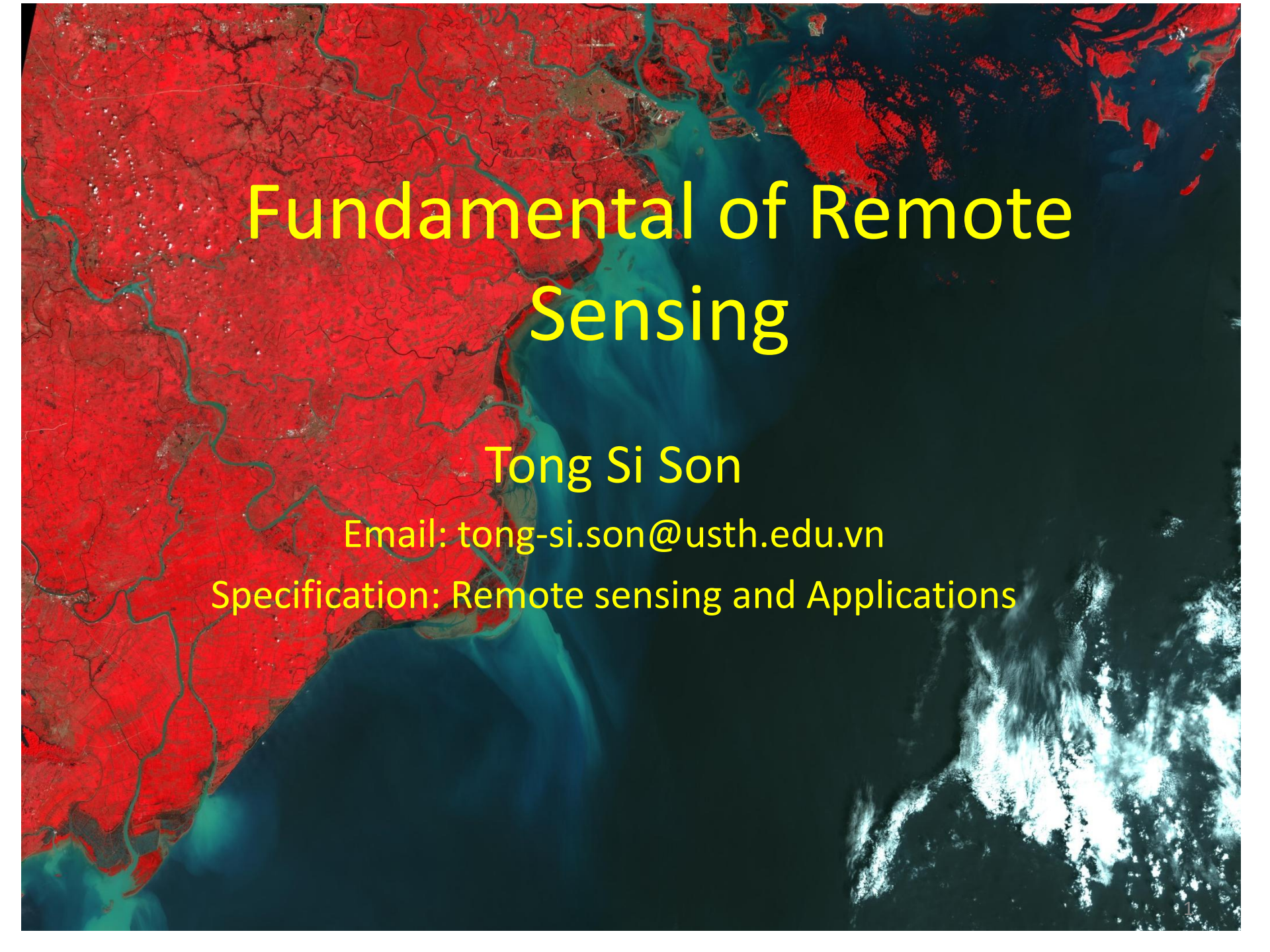


A satellite image of a coastal region. The land is colored red, showing a complex network of rivers and waterways. The surrounding water is dark blue. In the bottom right corner, there is a bright, white, cloud-like feature. The text "Fundamental of Remote Sensing" is overlaid in yellow.

Fundamental of Remote Sensing

Tong Si Son

Email: tong-si.son@usth.edu.vn

Specification: Remote sensing and Applications

Structure Of The Course

- **Time**

36 hours: Lecturer: 21 h, Practice: 15

- **Contents**

1. Overview of Remote sensing
2. Physical basis of Remote sensing
3. Structure of remote sensing systems
4. Satellite image characteristics
- 5. Satellite image pre-processing**
6. Satellite image classification and accuracy assessment

Section 5. Satellite image pre-processing

(Reference material from Prof. Gabriëlle De Lannoy, KU Leuven)

5.1 Sources of distortion

5.2 Geometric correction

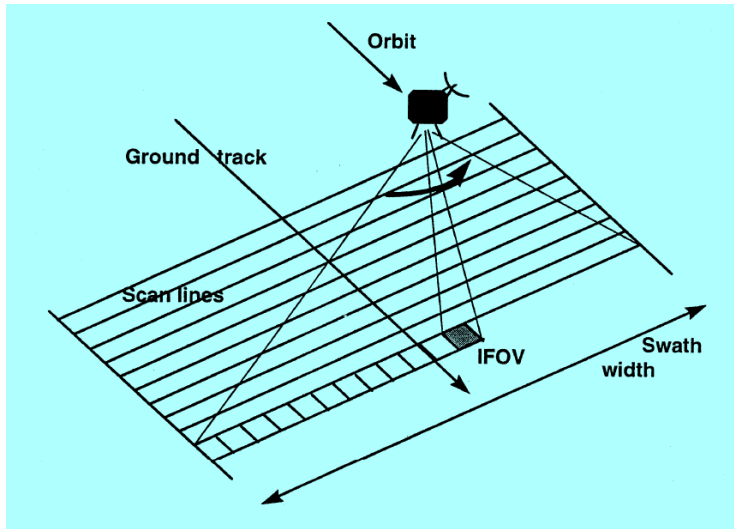
5.3 Radiometric calibration

5.4 Atmospheric correction

5.5 Radiometric correction

5.6 Image enhancement

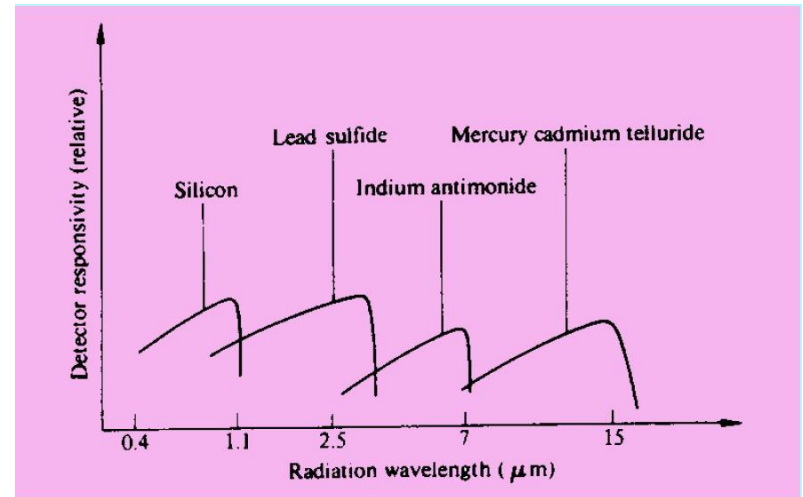
5.1 Sources of distortion



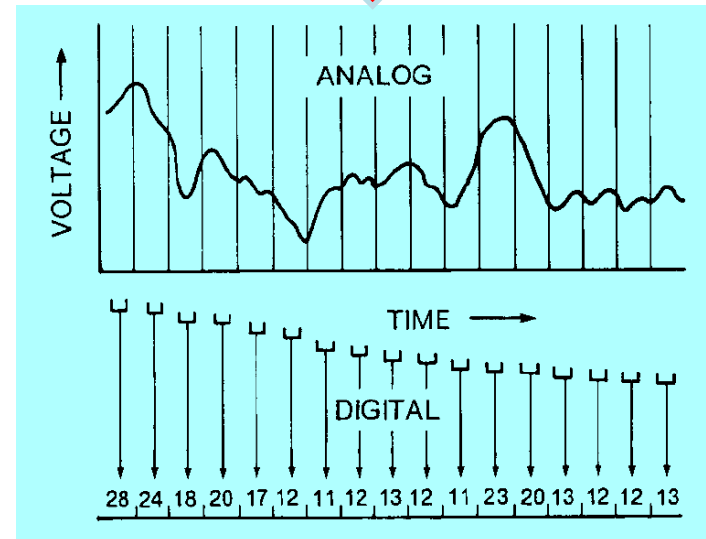
Acquiring images

119	122	113	113	125
121	124	115	113	125
122	125	123	118	122
120	116	118	123	121
117	121	120	125	133

Images



Sensitivity of sensor

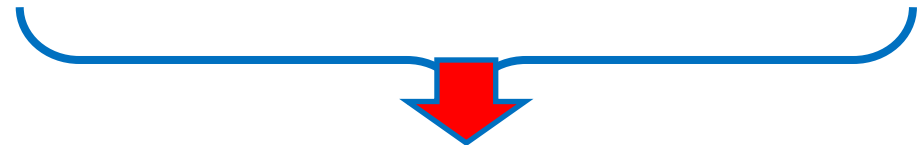


A/D conversion

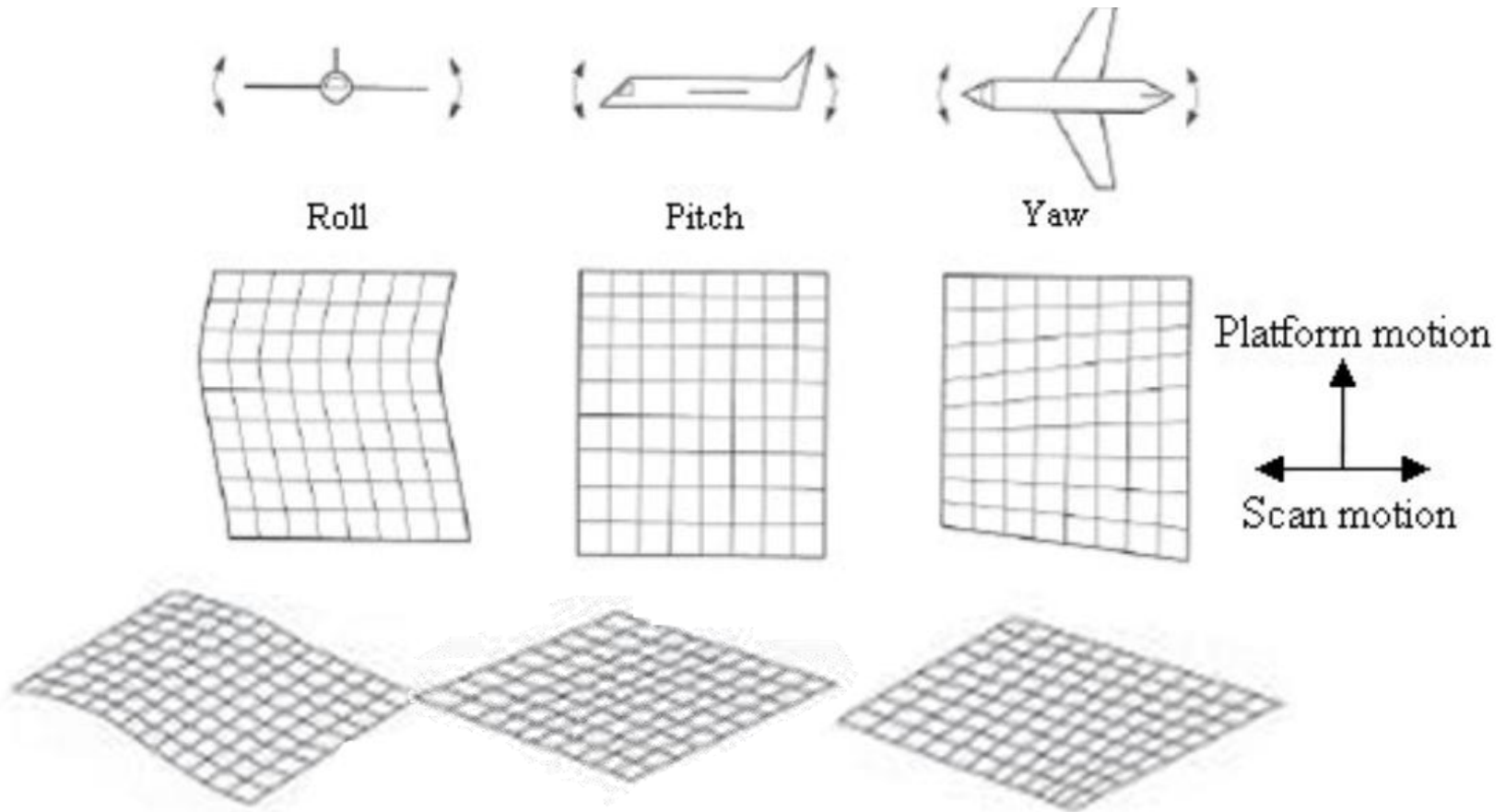
Sources of distortion

119	122	113	113	125
121	124	115	113	125
122	125	123	118	122
120	116	118	123	121
117	121	120	125	133

- 
1. Variations in flying height/orbit
 2. Panoramic distortions
 3. Relief displacement
 4. Sensor's non-uniform/ nonlinear behavior
 5. Variations in scene illumination/viewing geometry
 6. Atmospheric conditions
 7. Seasonal phenological variations

- 
1. Geometric correction
 2. Radiometric calibration
 3. Radiometric correction
 4. Atmospheric correction

Factor: Variations in flying height/orbit

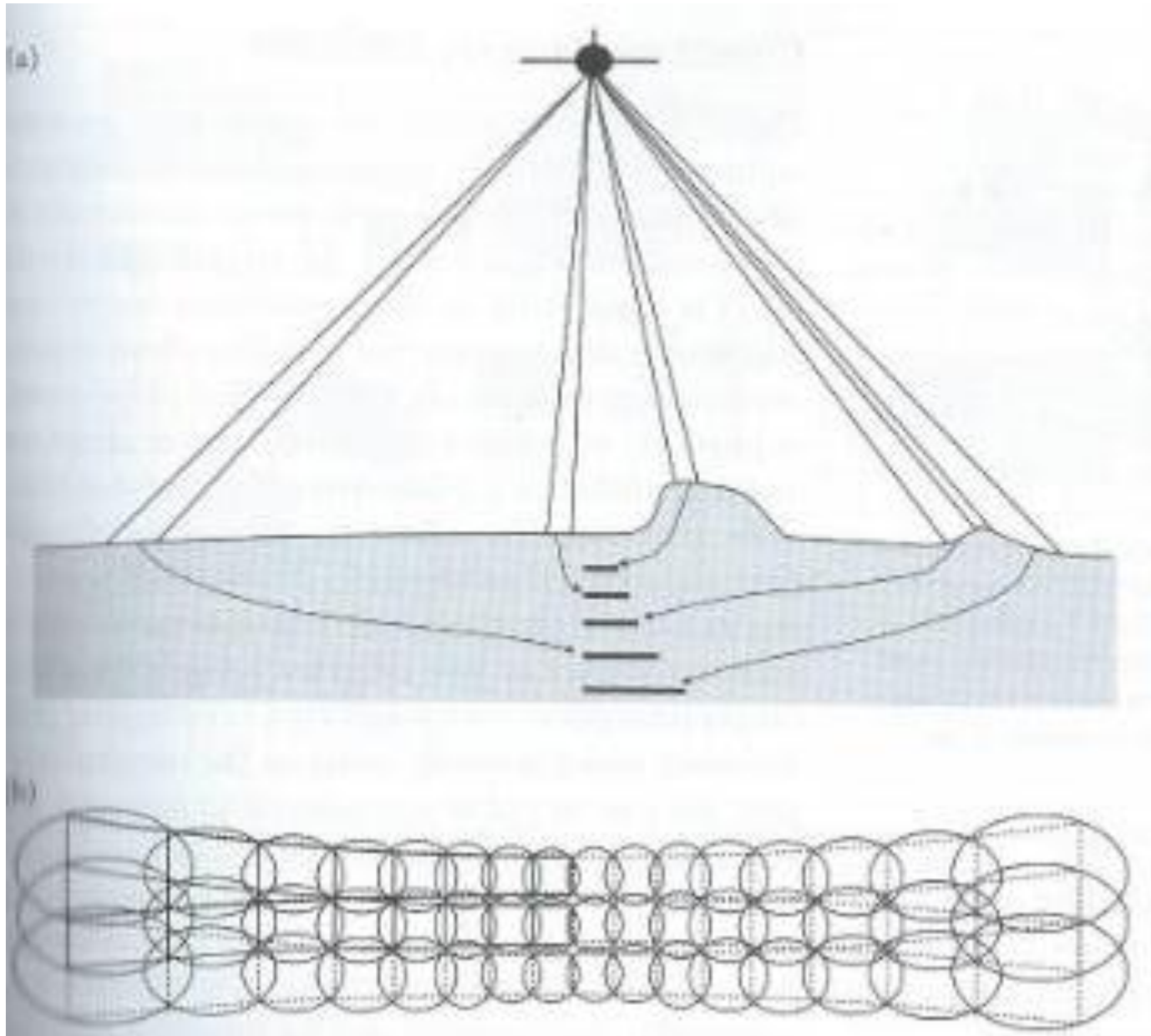


Roll: Variation in the swath position

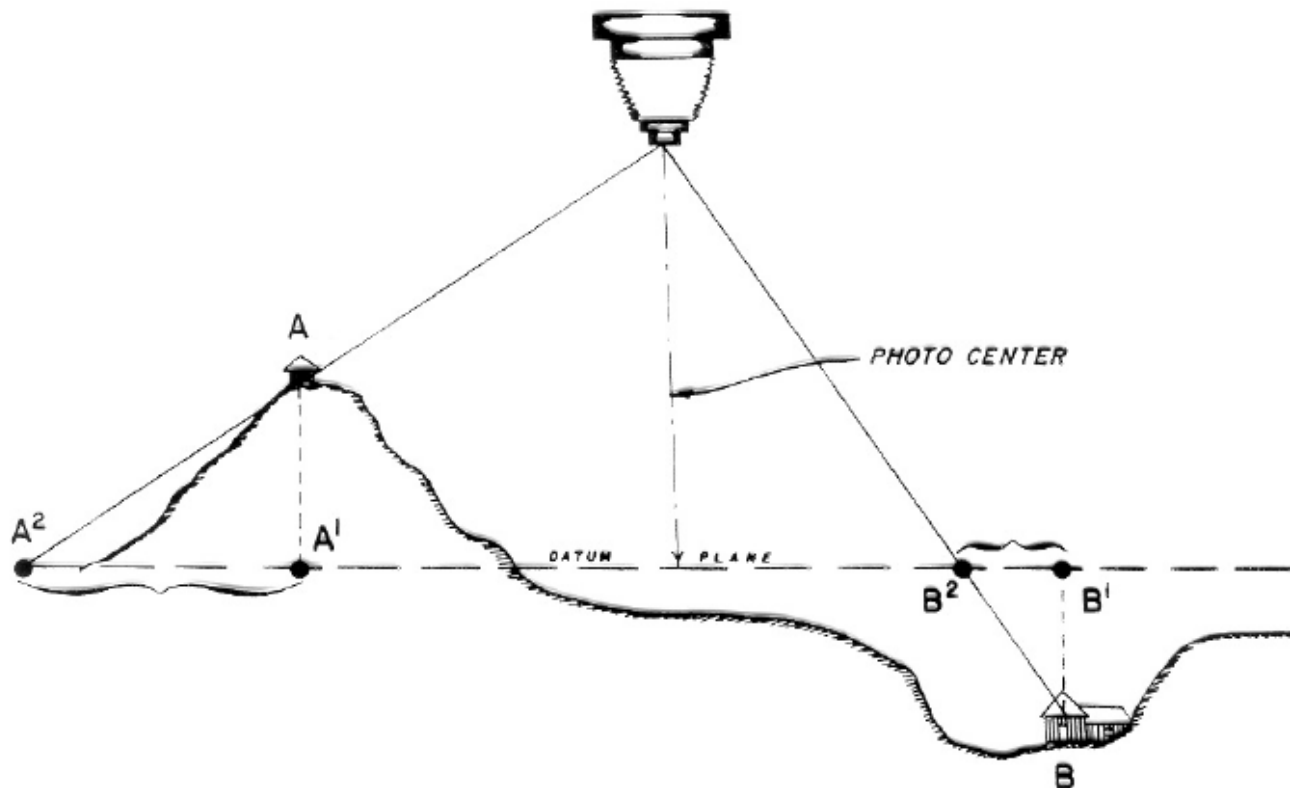
Pitch: Compression, expansion of whole scan lines

Yaw: Skewing of the scanlines

Factor: Panoramic distortions



Factor: Relief displacement



Ortho
rectification
using DEM

A' to A^2 = Distance of displacement away from center of photograph.

B' to B^2 = Distance of displacement towards center of photograph.

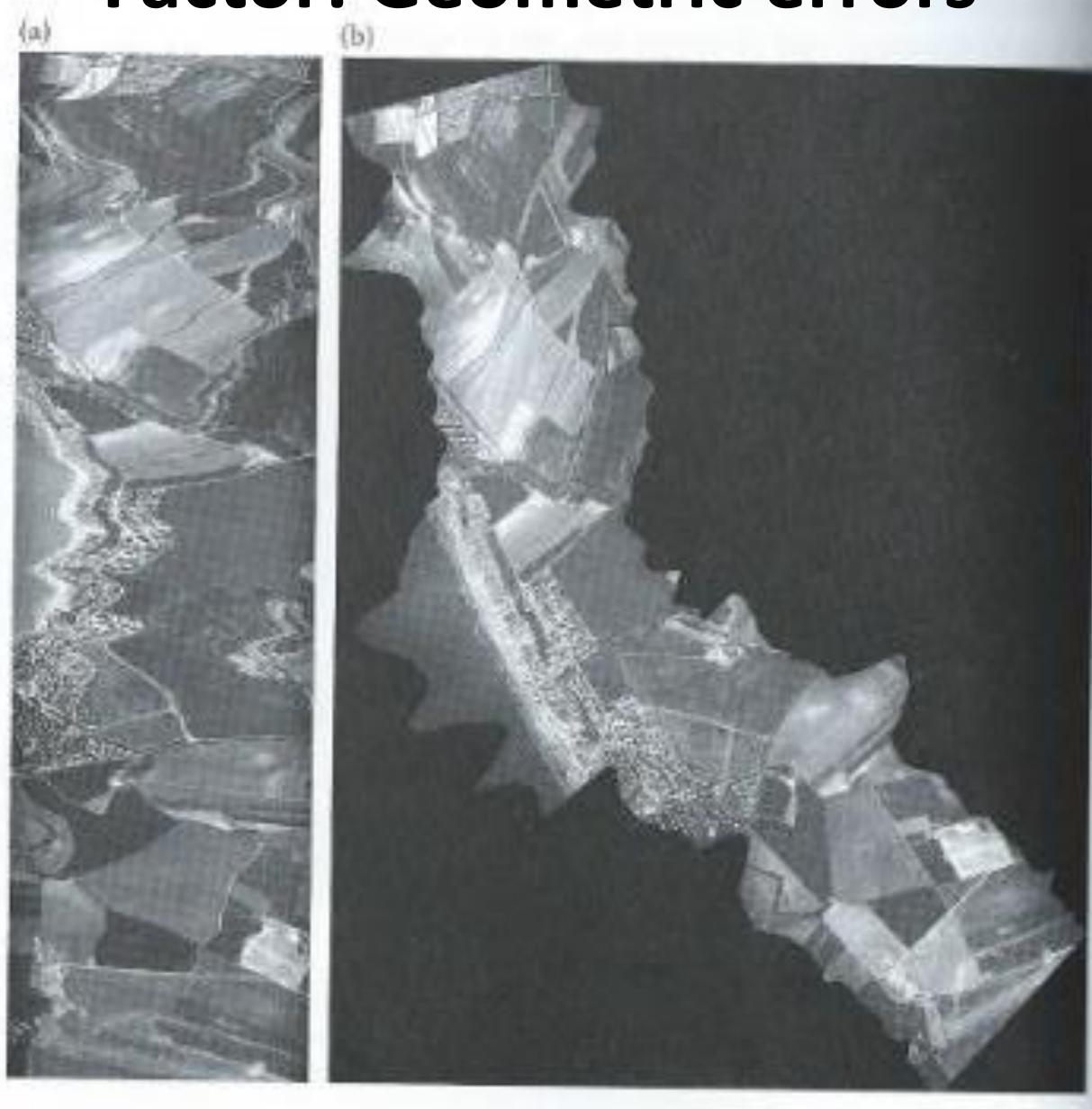
A' and B' = Corrected positions.

A^2 and B^2 = Displaced positions on photograph.

Factor: Relief displacement

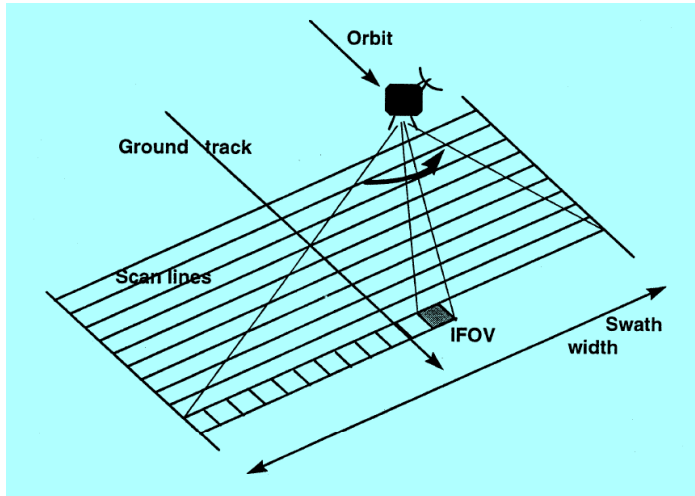


Factor: Geometric errors



Aircraft Instability

5.2 Geometric correction



119	122	113	113	125
121	124	115	113	125
122	125	123	118	122
120	116	118	123	121
117	121	120	125	133

Field measurements, GIS data



Geometric correction

- Image registration (so called Image rectification)

Image rectification

Removal of spatial sensor distortions and link it known geographic coordinate system

- Registration: matching two (or more) images to each other
- Link to geographic coordinate system: Using field measurements, GIS data

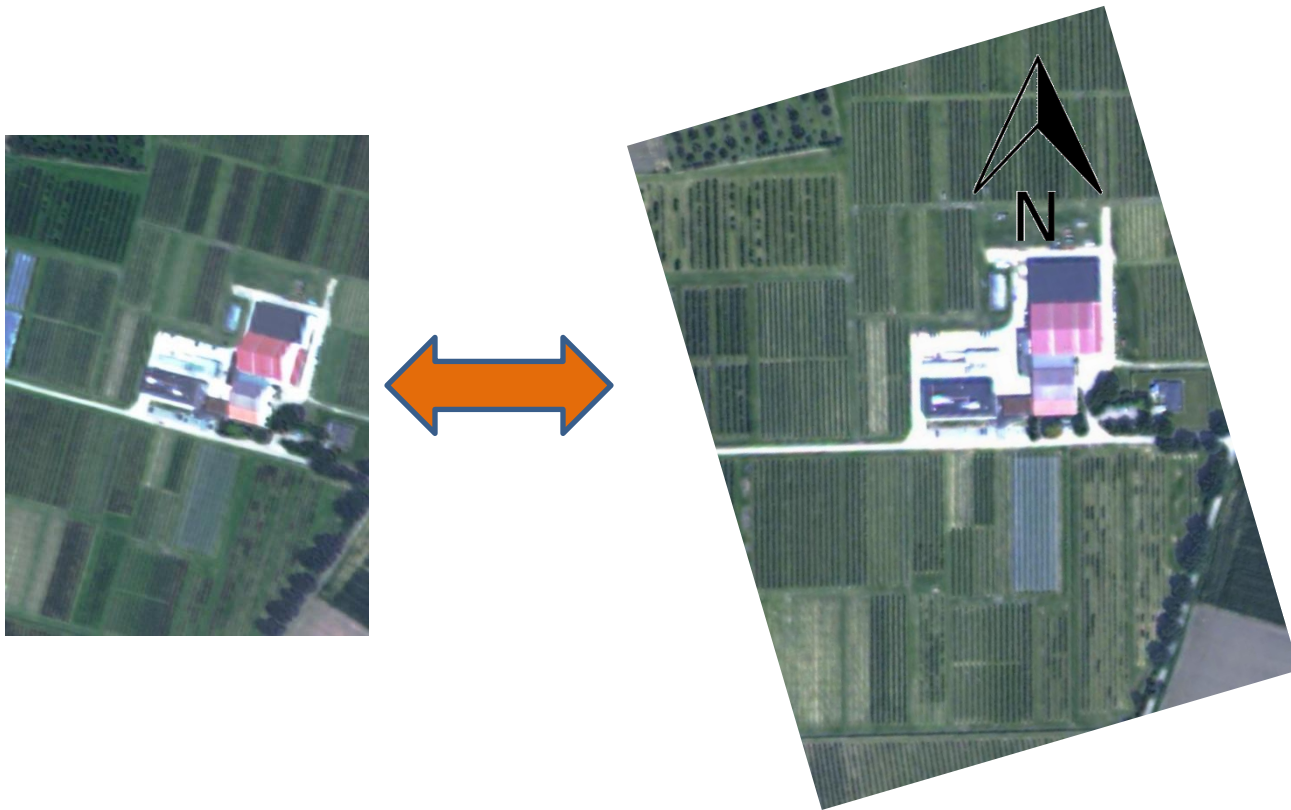


Image rectification

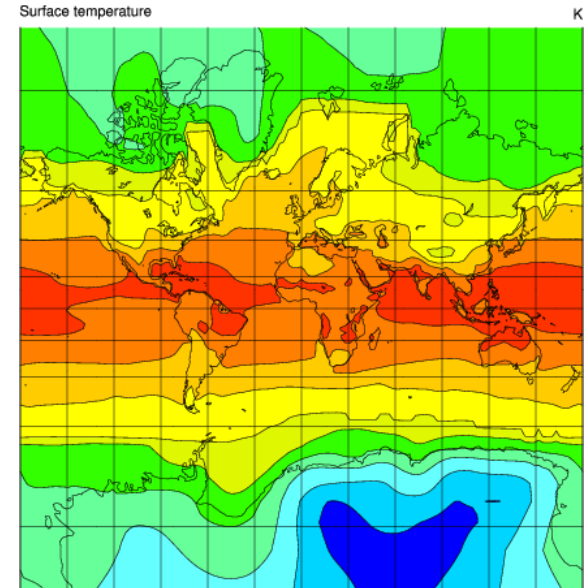
Convert image to map

Reference system = coordinate system

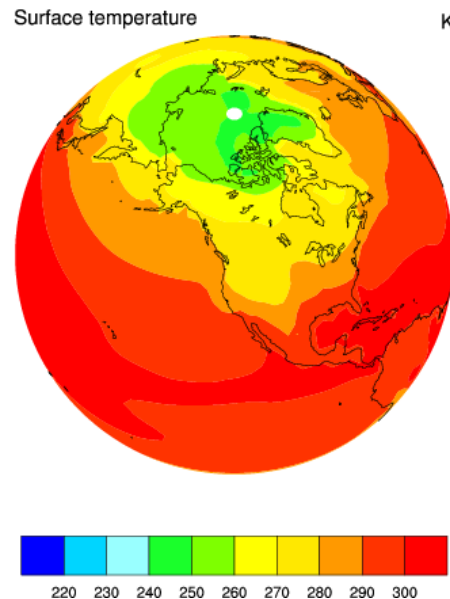
Geographic vs projected

- Using Image warping (*for plane area*)
- Using a DEM (*for mountainous area*)

Example of Mercator Projection



Example of Orthographic Projection



Example of Mollweide Projection

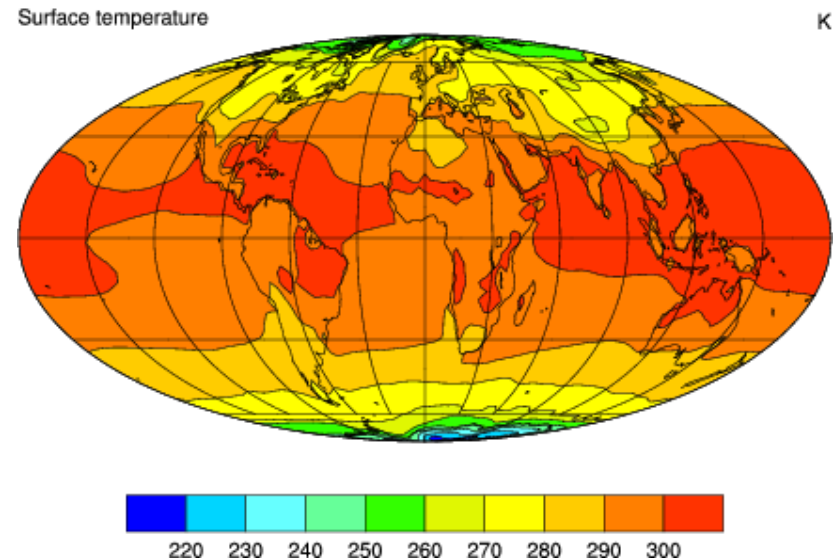


Image rectification

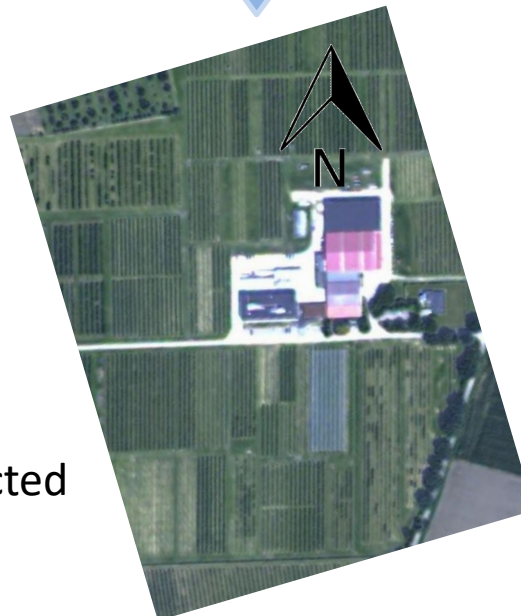
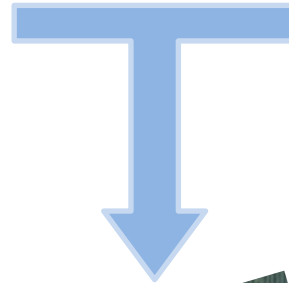
Using Ground control points (GCPs)

Properties of GCP network

- Stability (Available on the field)
- Extractability (Clearly observe in Image)
- Frequency (number of GCP)
- Evenly distributed
- Output coordinates known



Image warping: Plane area



Corrected
image

Image rectification

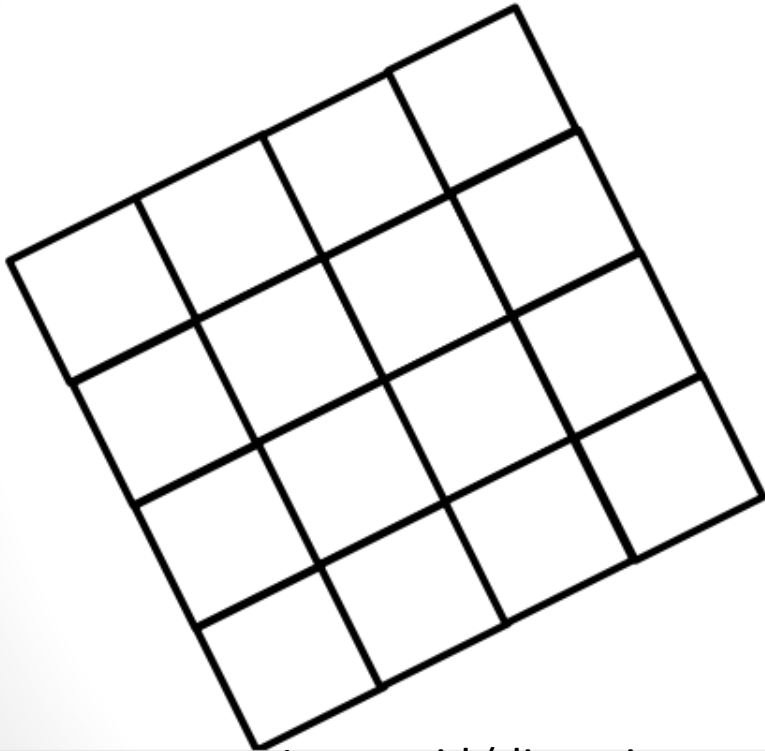
Two steps of image rectification

1/ Spatial interpolation

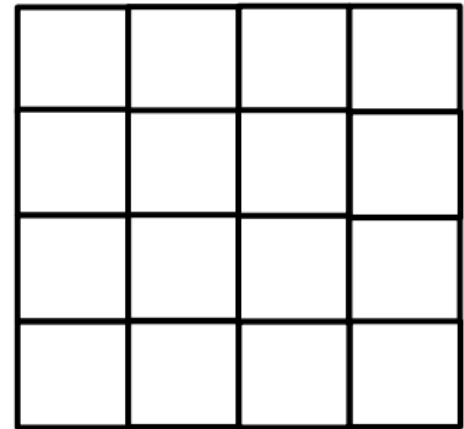
2/ Intensity interpolation (resampling)

Image rectification

1/Spatial interpolation



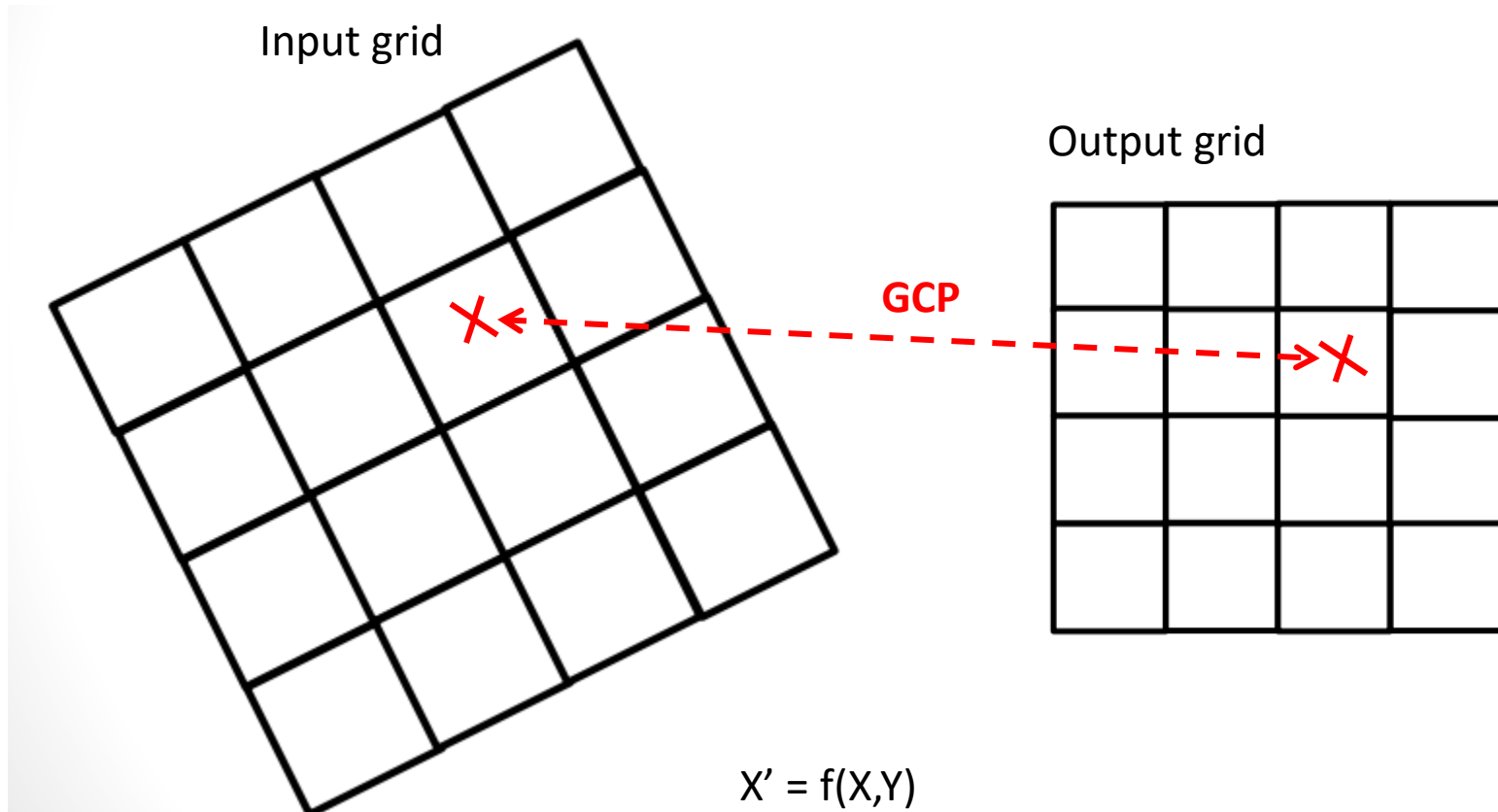
Input grid (distortion
and non referent
system)



Output grid (No
distortion, registered in
a coordinate system)

Image rectification

1/Spatial interpolation



$$X' = f(X,Y)$$

$$Y' = f(X,Y)$$

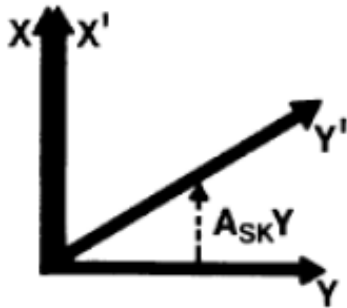
X', Y' : input coordinates

X, Y : output coordinates

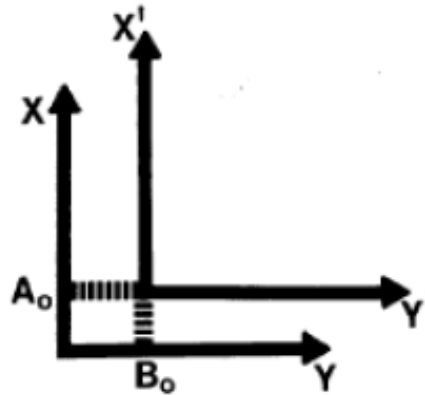
Image rectification

First order warp (W=1)

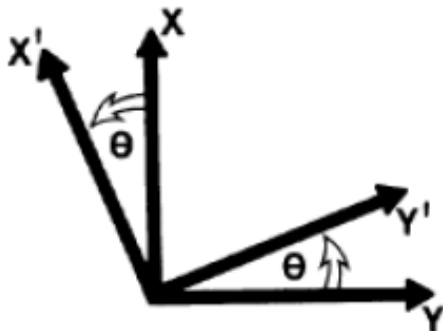
Skew
correction: $X' = X + A_{sk} Y$
 $Y' = Y$



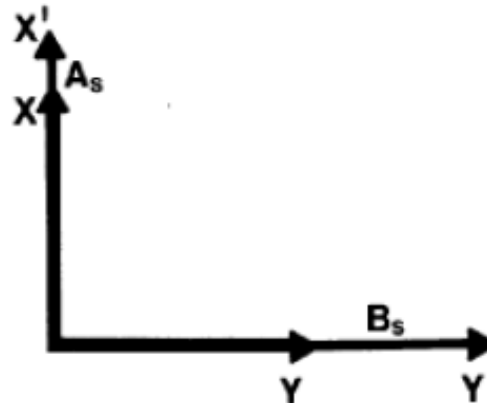
Translation
correction: $X' = A_o + X$
 $Y' = B_o + Y$



Rotation
correction: $X' = X \cos \theta - Y \sin \theta$
 $Y' = X \sin \theta + Y \cos \theta$



Scaling
correction: $X' = A_s X$
 $Y' = B_s Y$



$$X' = a_0 + a_1 X + a_2 Y$$

$$Y' = b_0 + b_1 X + b_2 Y$$

X, Y : Correct coordinates
 X', Y' : Distort coordinates

Image rectification

Euclidean / Procrustes	$\begin{aligned} x' &= ax + by + t_x \\ y' &= -bx + ay + t_y \end{aligned} \quad , \quad \begin{aligned} a &= s \cos \alpha \\ b &= s \sin \alpha \end{aligned}$
Affine / 1 st order polynomial	$\begin{aligned} x' &= a_0 + a_1x + a_2y \\ y' &= b_0 + b_1y + b_2y \end{aligned}$
Bilinear	$\begin{aligned} x' &= a_0 + a_1xy + a_2x + a_3y \\ y' &= b_0 + b_1xy + b_2x + b_3y \end{aligned}$
Perspective	$\begin{aligned} x' &= (a_0 + a_1x + a_2y)/(c_0x + c_1y + 1) \\ y' &= (b_0 + b_1x + b_2y)/(c_0x + c_1y + 1) \end{aligned}$
2 nd order polynomial / Biquadratic	$\begin{aligned} x' &= a_0 + a_1x + a_2y + a_3x^2 + a_4y^2 + a_5xy \\ y' &= b_0 + b_1x + b_2y + b_3x^2 + b_4y^2 + b_5xy \end{aligned}$
General polynomial	$\begin{aligned} x' &= \sum_i \sum_j a_{ij} x^i y^j \\ y' &= \sum_i \sum_j b_{ij} x^i y^j \end{aligned}$

$$\text{number of GCP's} = \frac{(W+1) \times (W+2)}{2}$$

W = order of equation (i.e. 1, 2, ...)

Image Rectification

- Accuracy assessment (Based on GCP's)
 - Number of GCP's dependent on rectification method + terrain
 - Selection of GCP's
- Residual = Coordinates of GCP on image after correction – Coordinates of GCP

GCP #	Image X pixel	X residual x	Image Y pixel	Y residual
1	1269.75	-0.24	1247.59	0.13E+02
2	867.91	-0.61E+01	1303.90	0.89E+01
3	467.79	-0.11E+02	1360.51	0.55E+01
4	150.52	0.68E+02	1413.42	-0.86E+01
5	82.20	-0.38E+01	163.19	0.61E+01
6	260.89	0.28E+01	134.23	0.52E+01
7	680.59	0.35E+01	70.16	0.92E+01
8	781.85	-0.29E+02	714.94	-0.15E+03
	X rms error	18.26	Y rms error	35.33
Overall rms error	39.77			

Image rectification

2/Intensity interpolation (Resampling)

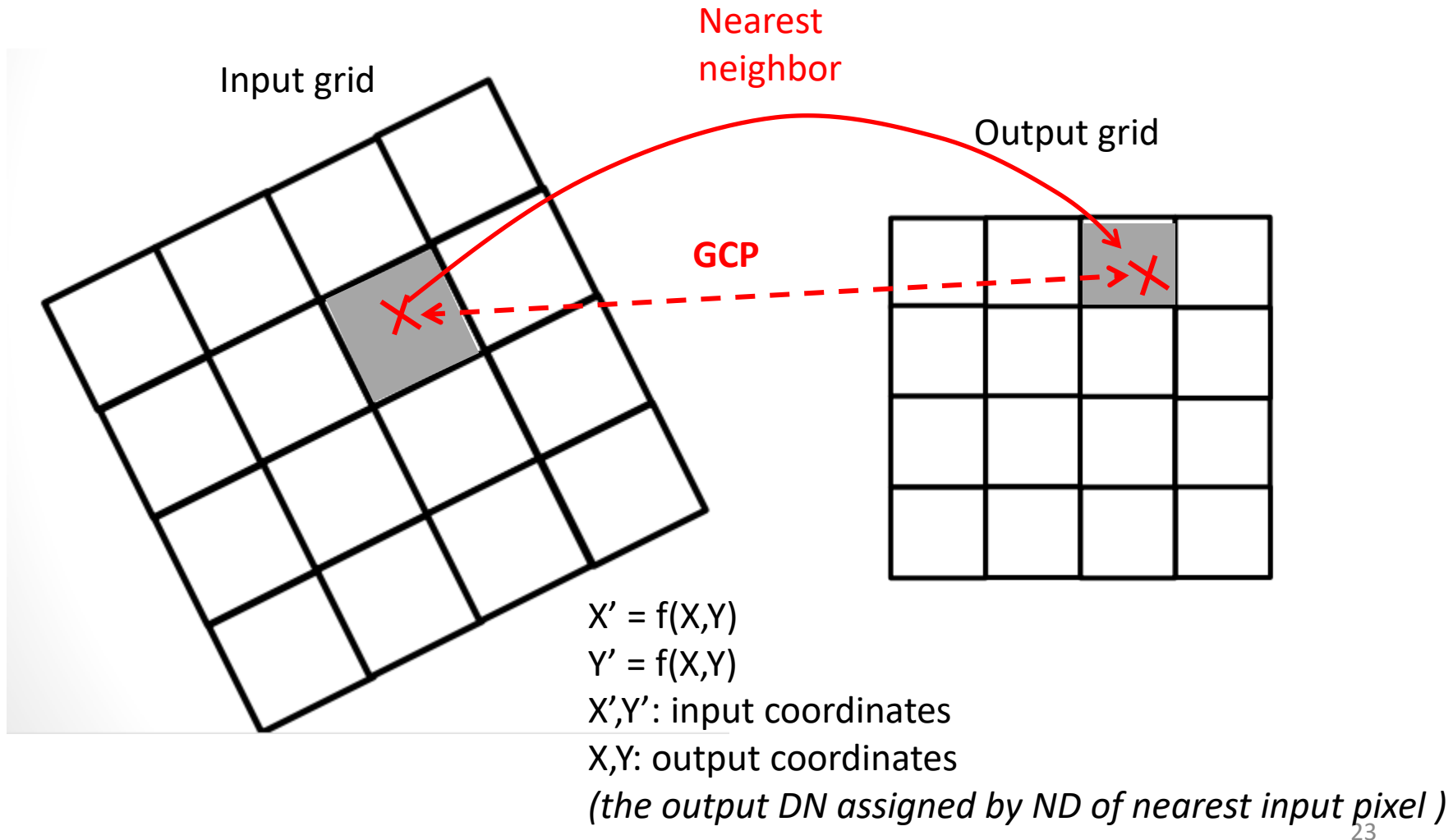


Image rectification

2/Intensity interpolation

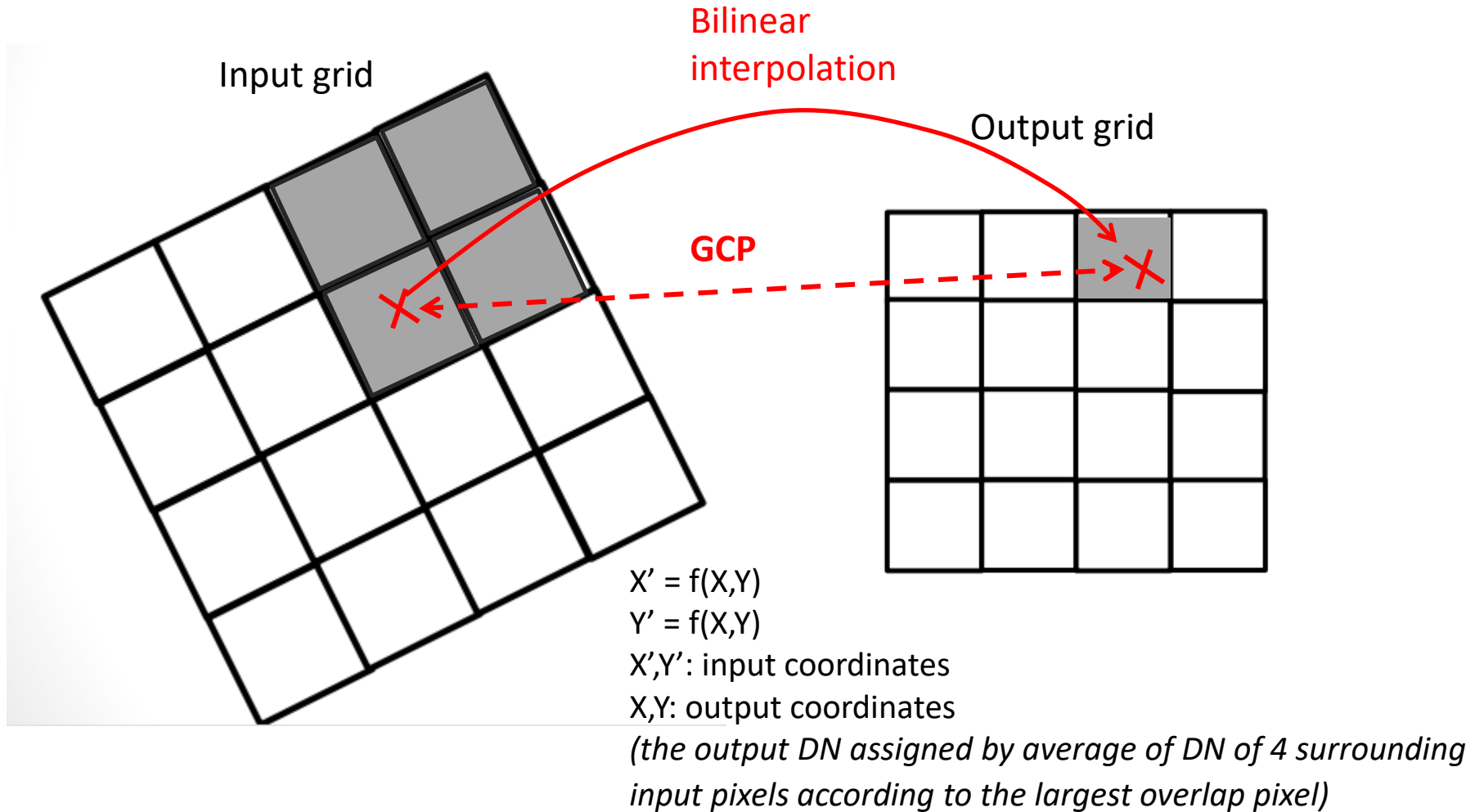
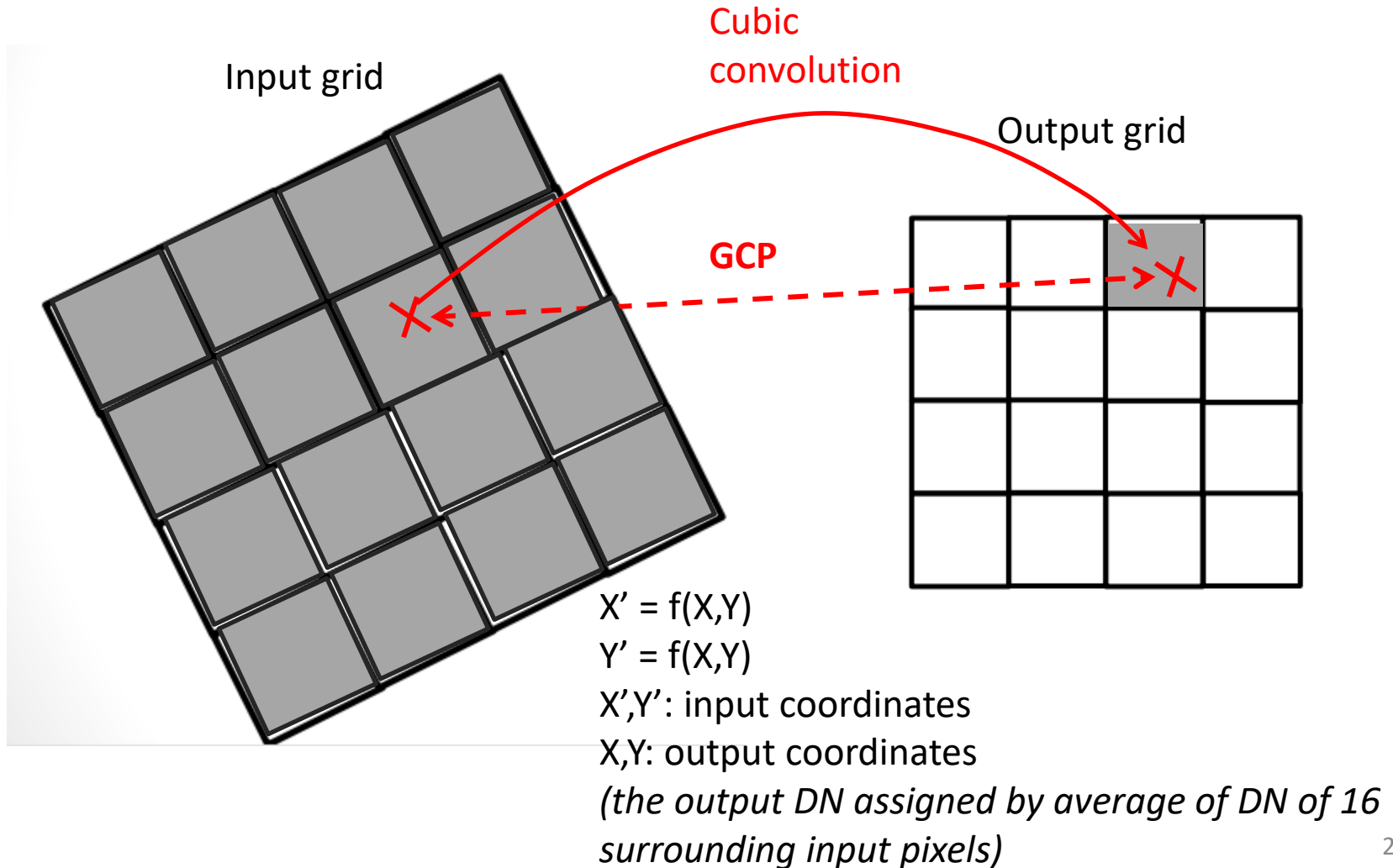
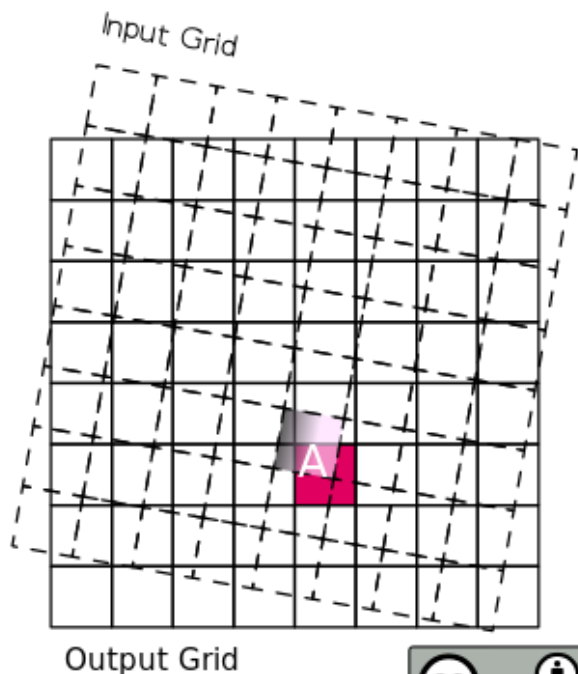


Image rectification

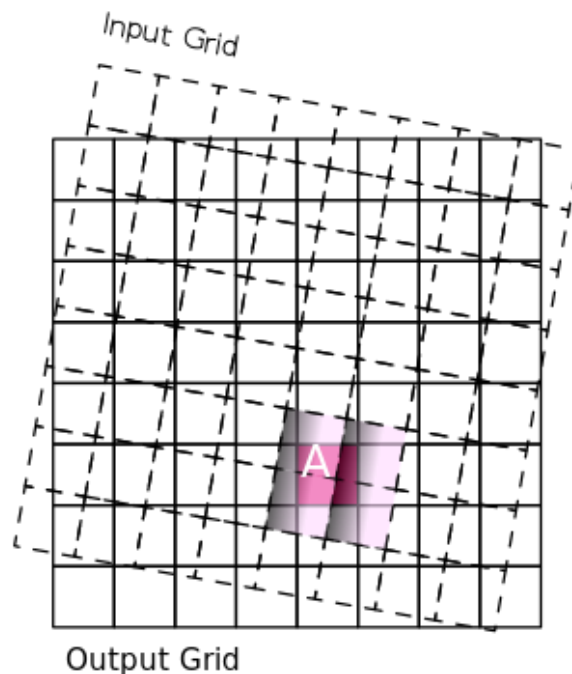
2/Intensity interpolation



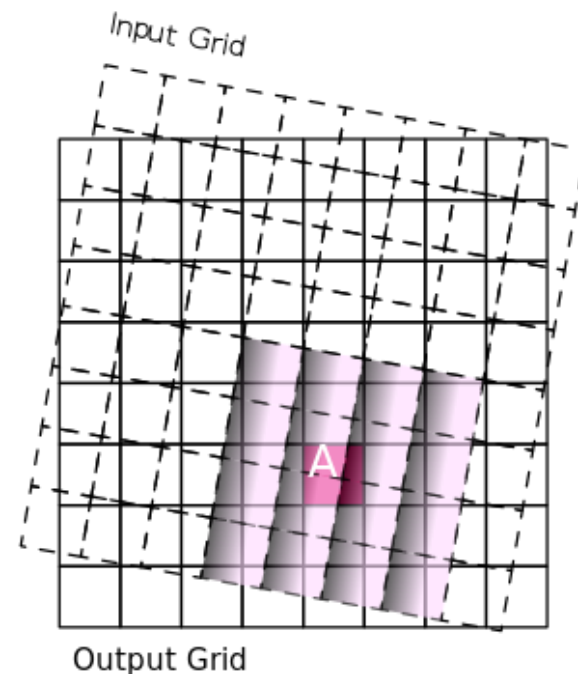
Nearest Neighbour



Bilinear Interpolation



Cubic Convolution



This work is licensed under a Creative Commons Attribution 3.0 Unported License.
Author: <http://commons.wikimedia.org/wiki/User:Arbeck>

Image rectification

2/Intensity interpolation
(for the same resolution)

1	3	6	1
3	3	1	7
8	1	8	7
6	1	1	1

1

(Original image)

1	3	6	1
3	3	1	7
8	1	8	7
6	1	1	1

2

(Nearest neighbour)

3	3	4	4
4	3	6	6
4	3	4	4
4	2	3	4

3

(Cubic convolution)

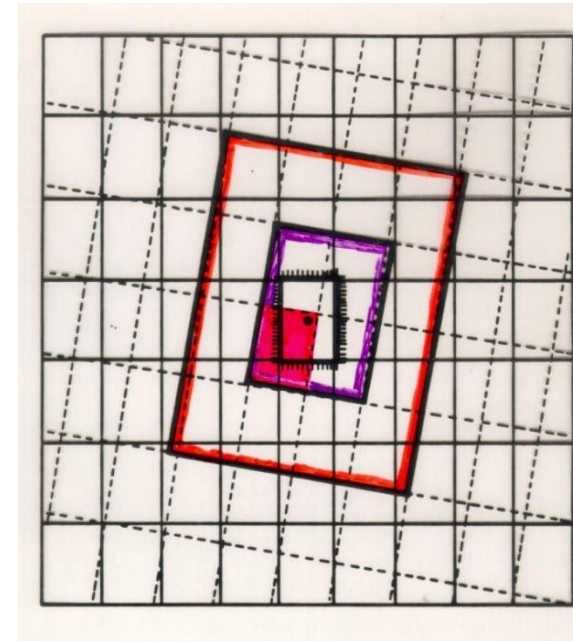
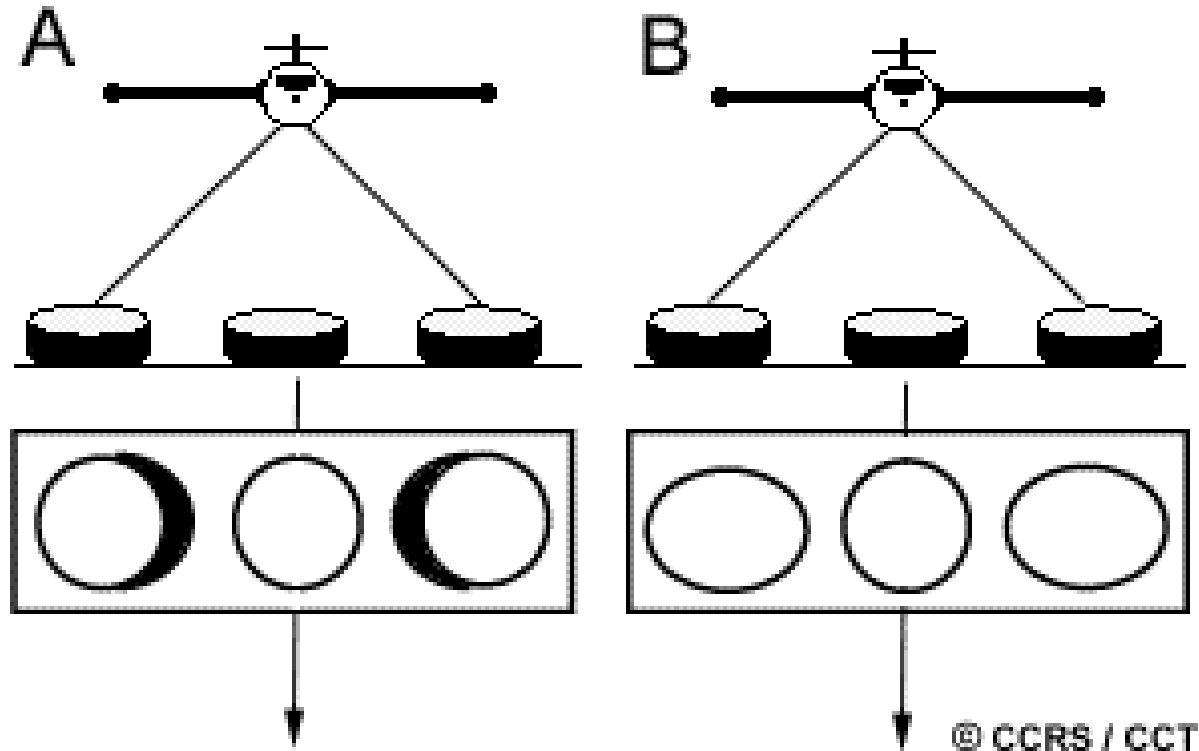


Image rectification (Orthorectification)

Based on Digital Elevation Model (DEM)

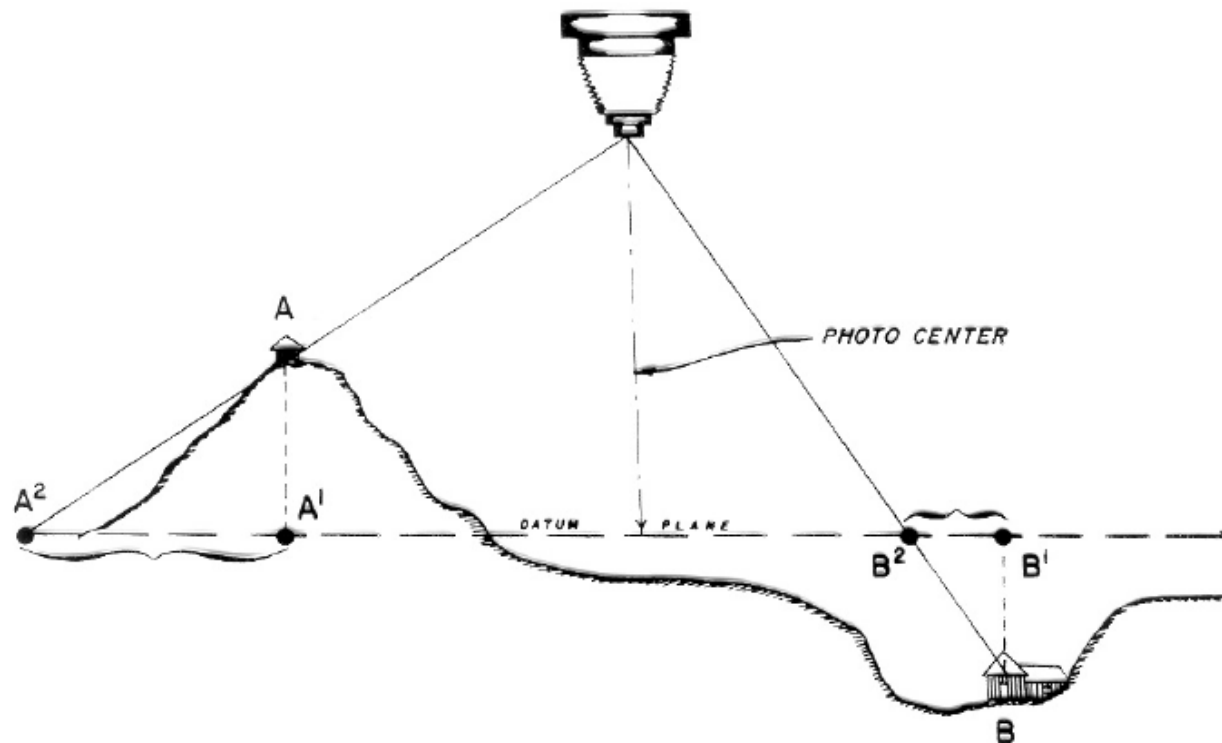


Relief displacement

Scale distortion

Image Rectification

Based on Digital Elevation Model (DEM)



A^1 to A^2 = Distance of displacement away from center of photograph.

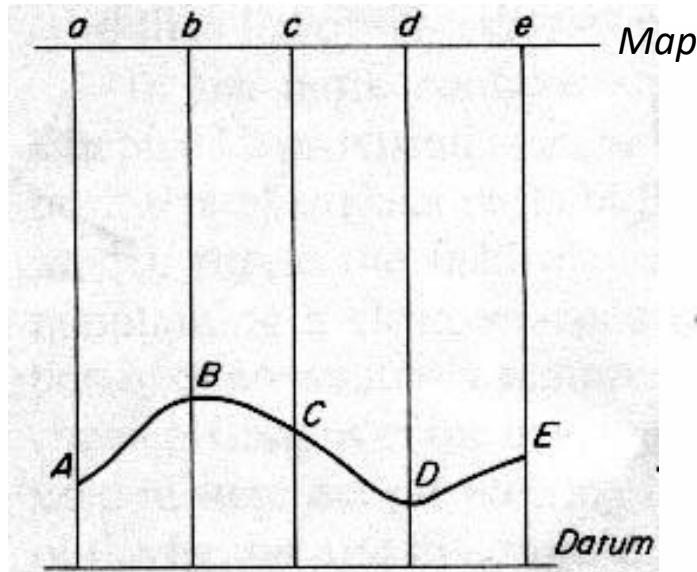
B^1 to B^2 = Distance of displacement towards center of photograph.

A^1 and B^1 = Corrected positions.

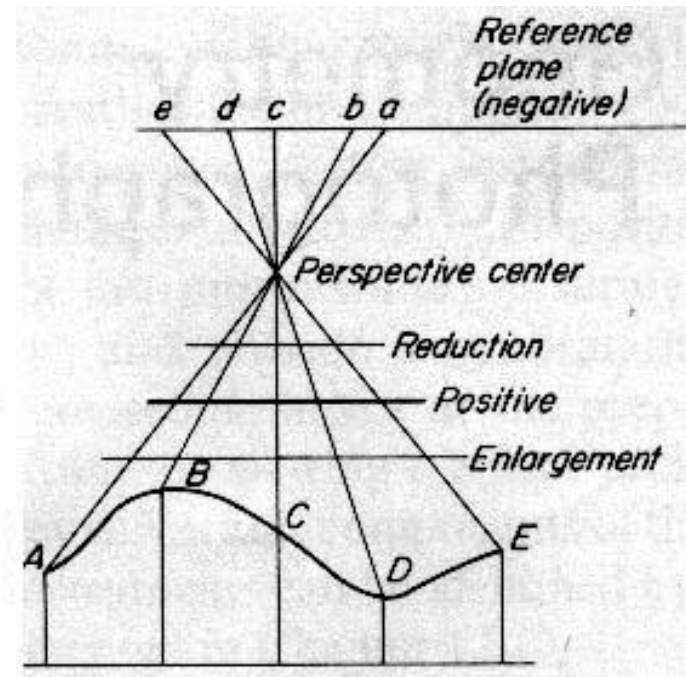
A^2 and B^2 = Displaced positions on photograph.

Image Rectification

Orthogonal vs. perspective projections



- Orthogonal projection.
- Uniform scale.
- No relief displacement.



- Perspective projection.
- Non-uniform scale.
- Relief displacement.

Image Rectification

Based on Digital Elevation Model (DEM)

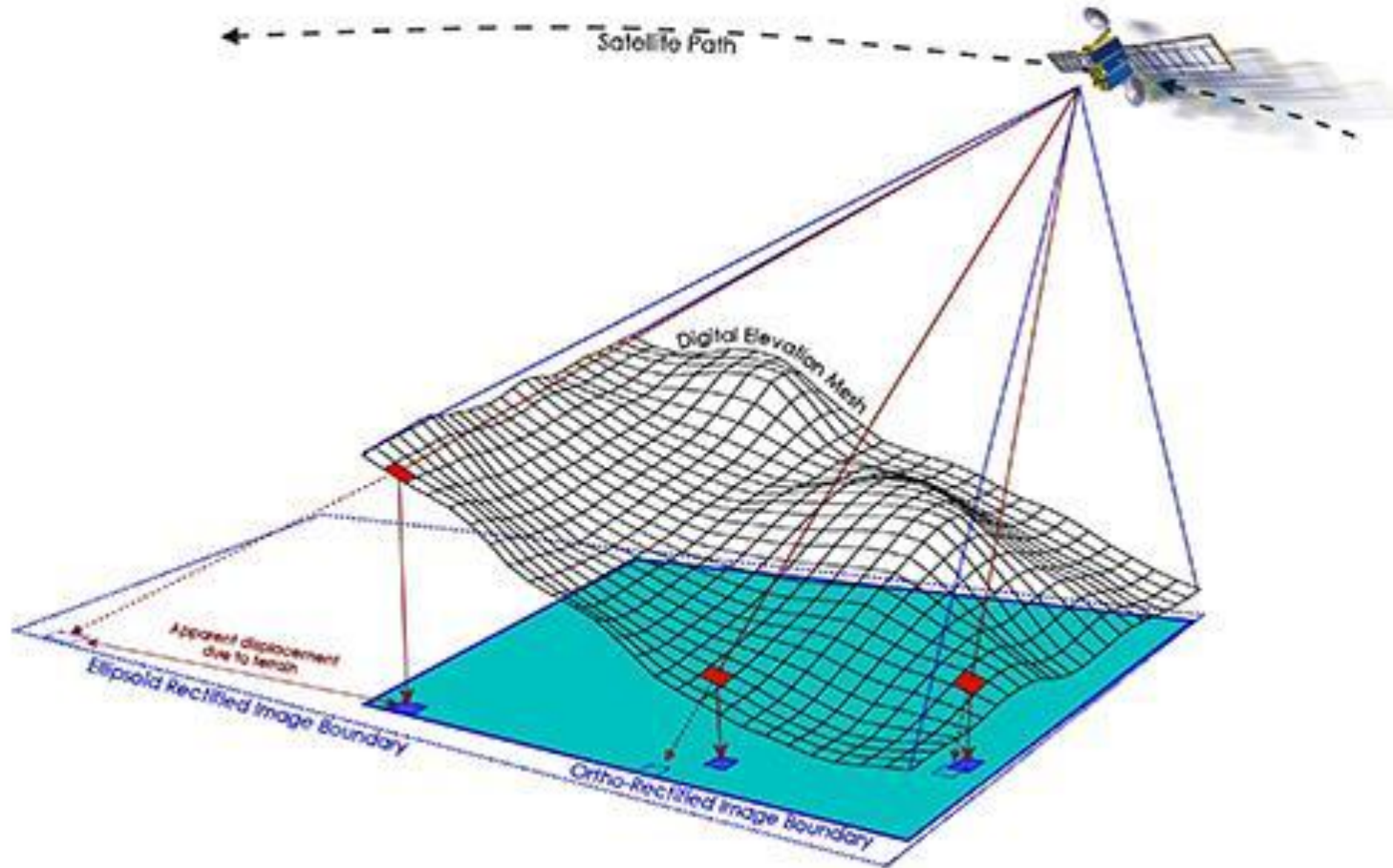
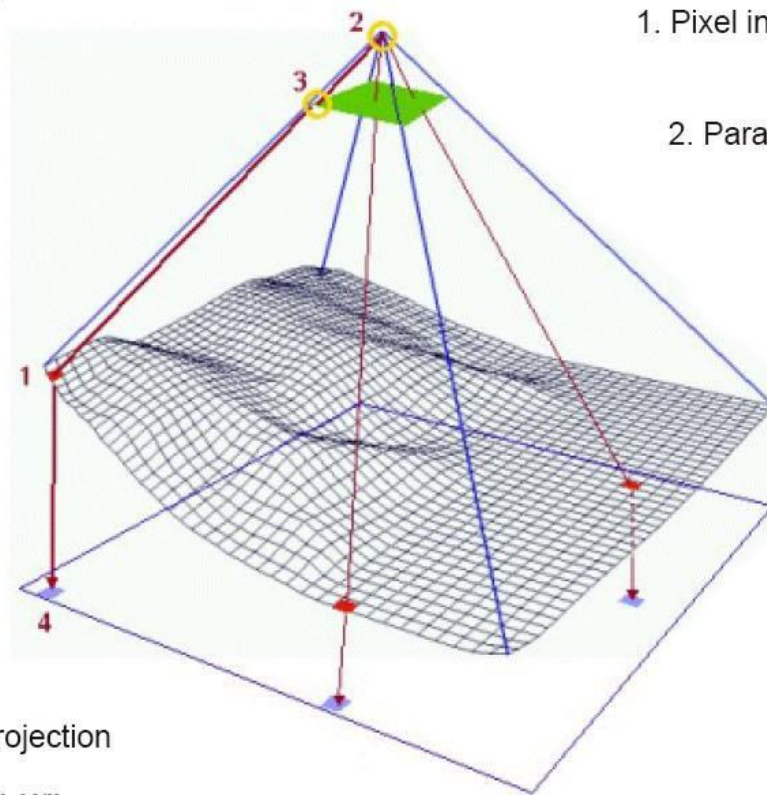


Image rectification

Based on Digital Elevation Model (DEM)

Orthorectification process of remote sensed Image data



1. Pixel in the DEM (Height)

2. Parameters in the Exterior Orientation

3. In the image, a Brightness Value is determined based on the resampling of surrounding pixels

4. Height, Exterior Orientation information and Brightness Value used to calculate equivalent location in the orthoimage

Orthographic Projection

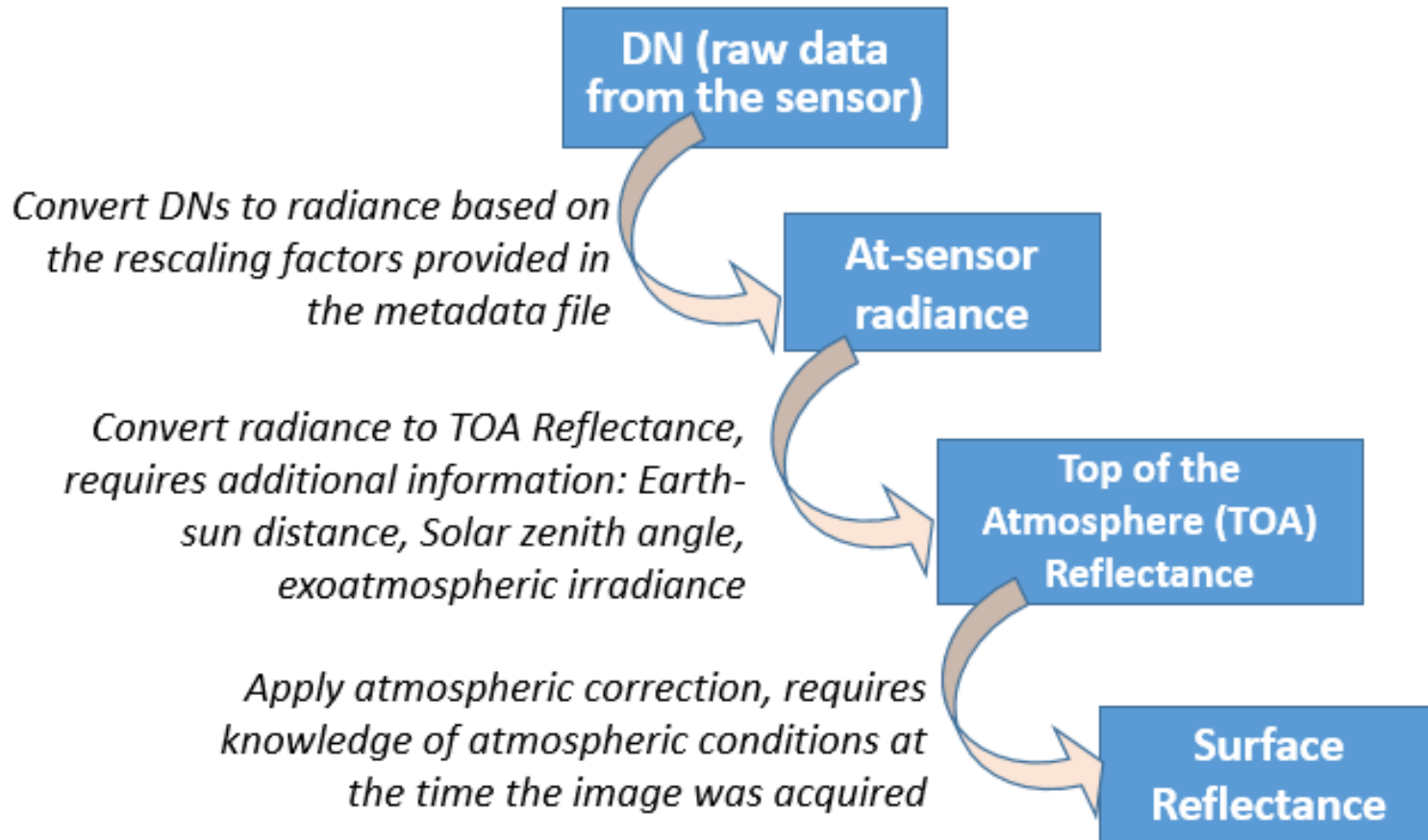
www.satimagingcorp.com

Copyright (c) 2005 Leica Geosystems - All rights reserved

Novak, K., (1992). "Rectification of digital imagery". Photogrammetric Engineering and Remote Sensing, 58(3): 339-344.

5.3 Radiometric calibration

Radiometric Calibration & Correction Process



Radiometric calibration

- Radiometric calibration = convert DN's to physical meaningful units (radiance, reflectance)

Time 1 or band 1

119	122	113	113	125
121	124	115	113	125
122	125	123	118	122
120	116	118	123	121
117	121	120	125	133



DN = 123

Rad = X [W/m² Sr μm]

Time 2 or band 2

119	122	113	113	125
121	124	115	113	125
122	125	123	118	122
120	116	118	123	121
117	121	120	125	133



DN = 123

Rad = 5 X [W/m² Sr μm]

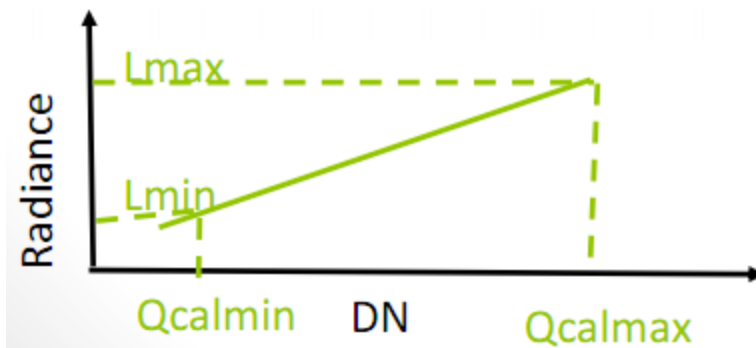
*(DN = 123 in both image,
Are they the same meaning?)*

Radiometric calibration

- Radiometric calibration = convert DN's to physical meaningful units (radiance, reflectance)
- At sensor radiance

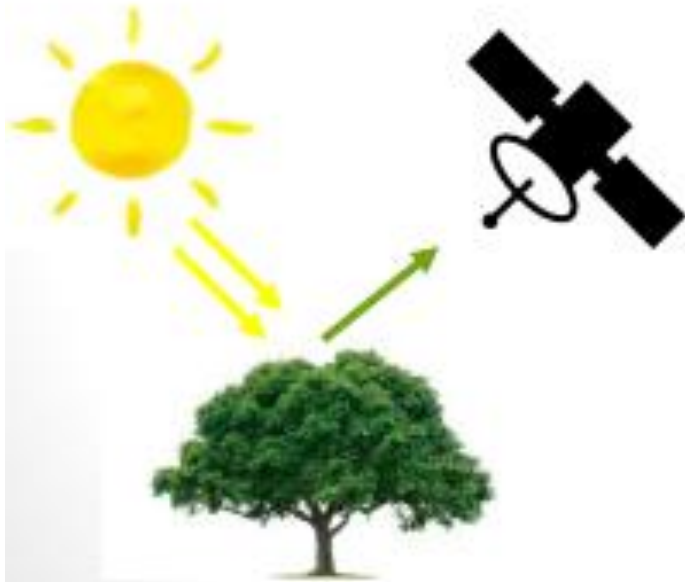
$$L_{\lambda} = LMIN_{\lambda} + \frac{LMAX_{\lambda} - LMIN_{\lambda}}{QCALMAX} \times QCAL$$

where QCAL =	Calibrated and quantized scaled radiance in units of DN
LMIN _λ =	Spectral radiance at QCAL = 0
LMAX _λ =	Spectral radiance at QCAL = QCALMAX
QCALMAX =	Range of re-scaled radiance in DN
L _λ =	Spectral radiance

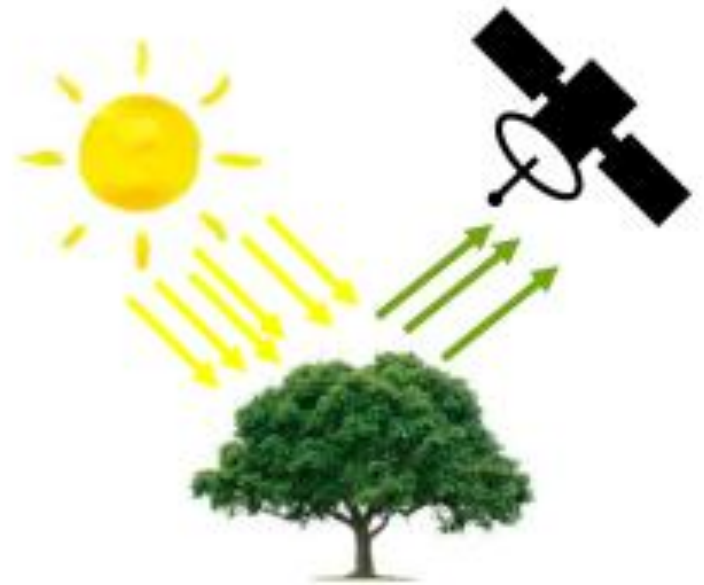


Radiometric calibration

- Convert DN's to physical meaningful units (radiance, reflectance)
- At sensor reflectance



Refl.: 50 [%]
Rad.: = X [W/m² Sr μm]



Refl.: 50 [%]
Rad.: = 3*X [W/m² Sr μm]

Radiometric calibration

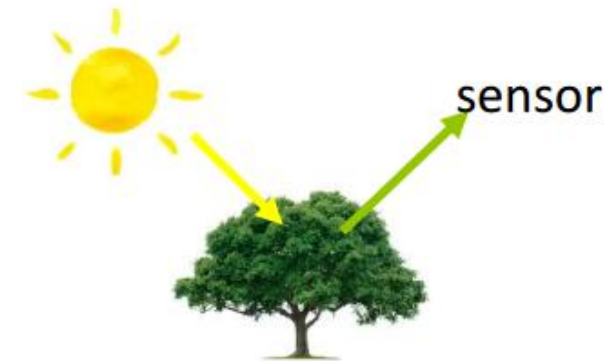
- Convert DN's to physical meaningful units (radiance, reflectance)
- At sensor reflectance

$$\rho = \frac{\pi \times L_{\lambda} \times d^2}{ESUN_{\lambda} \times \cos\theta_s}$$

where $\rho =$	Unitless effective at-satellite planetary reflectance
$L_{\lambda} =$	Spectral radiance at sensor aperture in $\text{mW} \times \text{cm}^{-2} \times \text{ster}^{-1}$ the previous equation
$d =$	Earth-sun distance in astronomical units
$ESUN_{\lambda} =$	Mean solar exo-atmospheric irradiance in $\text{mW} \times \text{cm}^{-2} \times \mu\text{m}^{-1}$
$\theta_s =$	Solar zenith angle in degrees

Radiometric calibration

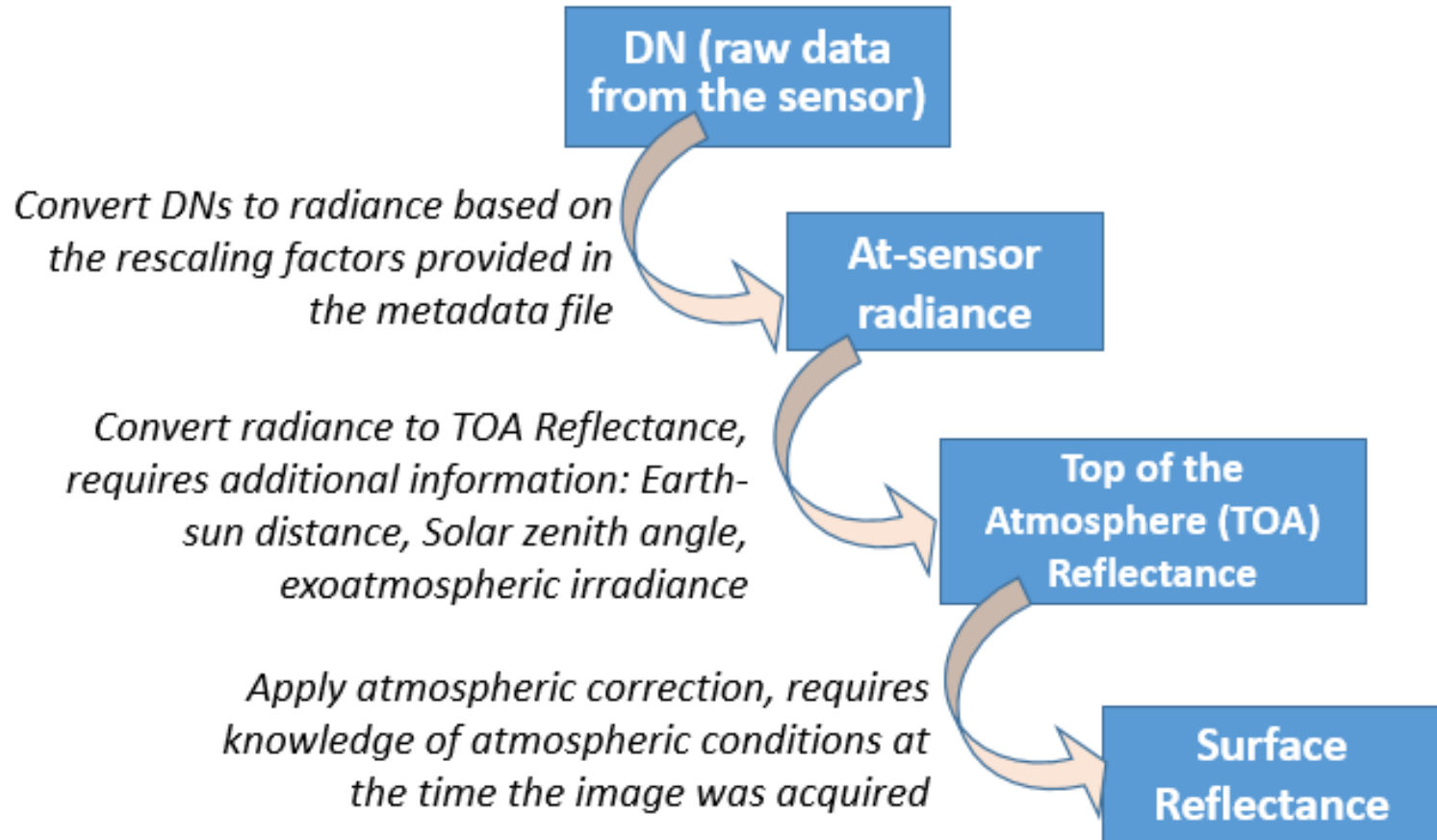
- Convert DN's to physical meaningful units (radiance, reflectance)
- At sensor reflectance



$$\text{Refl} = \frac{\text{refl rad}}{\text{incoming rad}}$$

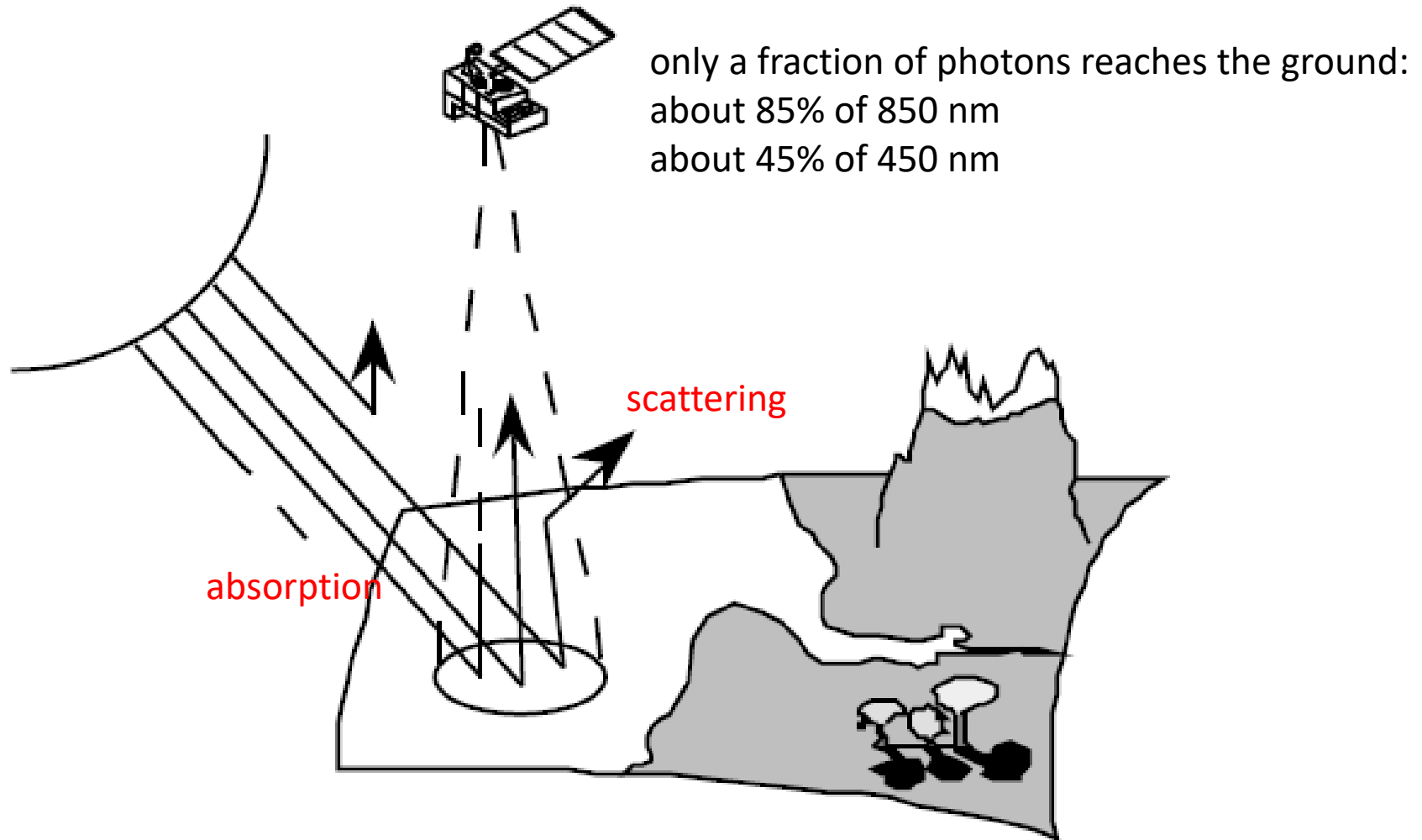
Radiometric correction and calibration

Radiometric Calibration & Correction Process



5.4 Atmospheric correction

Most importantly: atmospheric influence



Atmosphere correction

Most importantly: atmospheric influence

Absorption

- aerosols & gasses:

O₂
CO₂
CH₄
N₂O
..



relatively constant

O₃
H₂O

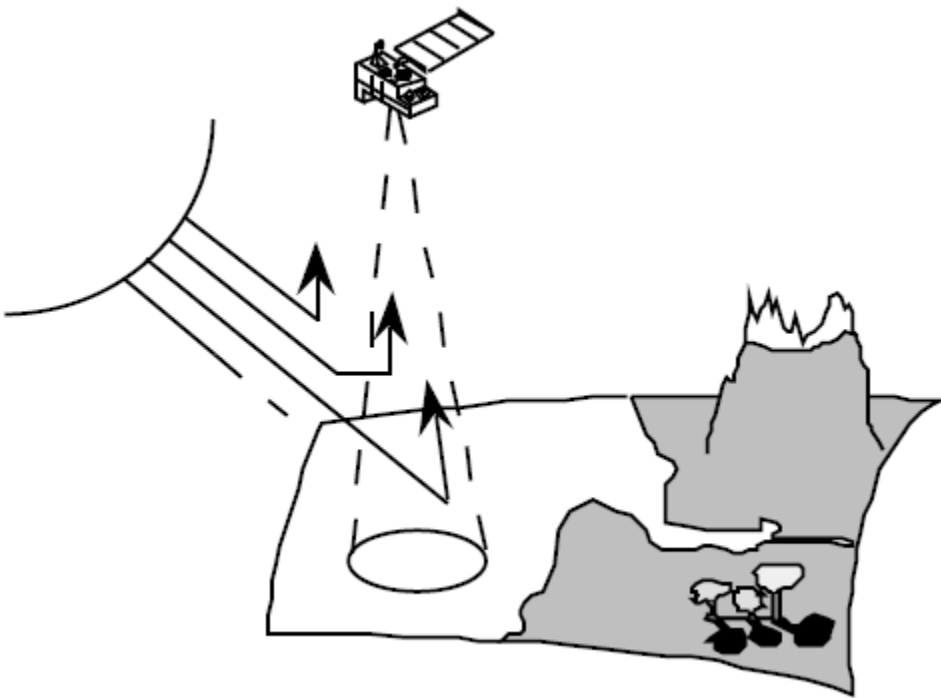


depend on time and location

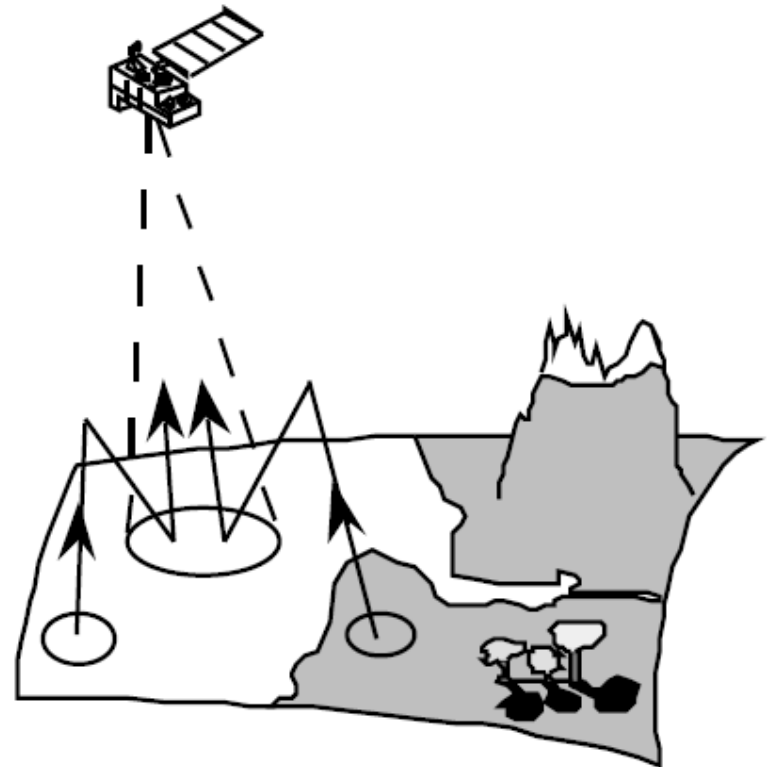
Atmosphere correction

Most importantly: atmospheric influence

Backscatter



Trapping effect

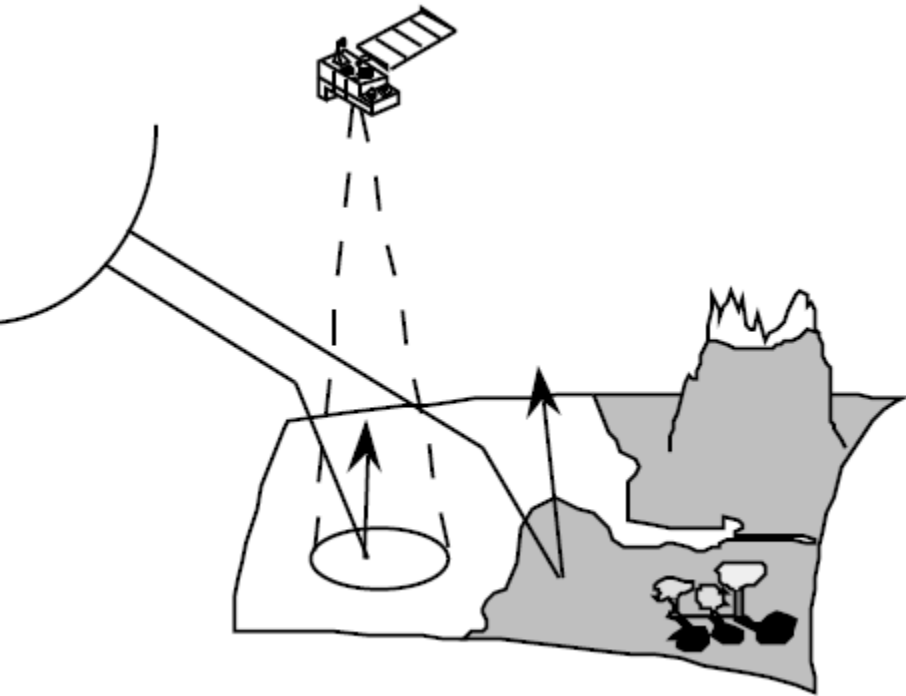


Atmosphere correction

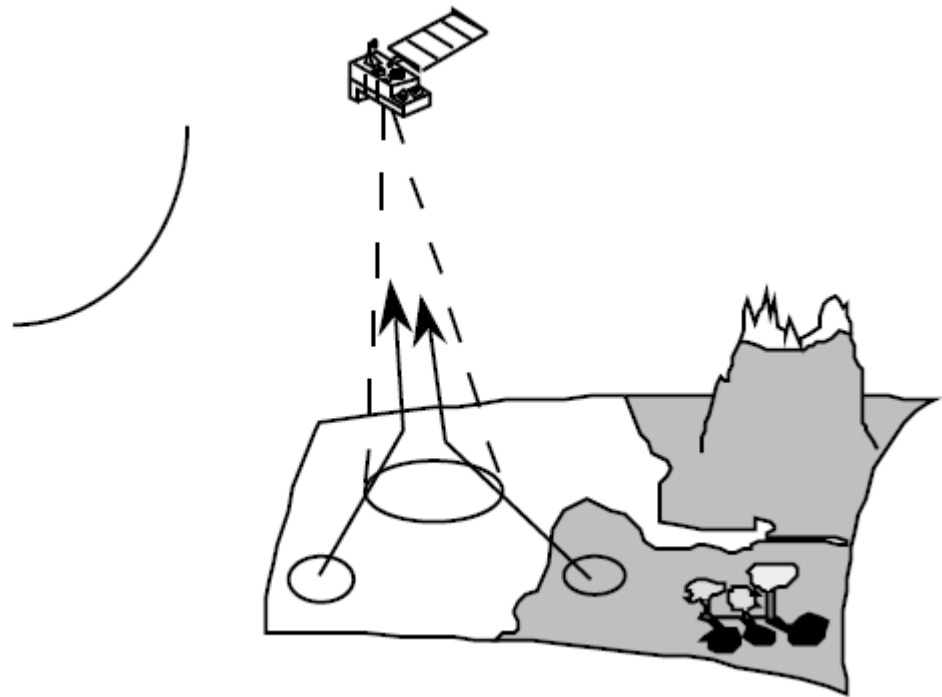
Most importantly: atmospheric influence

Scattering: path radiance

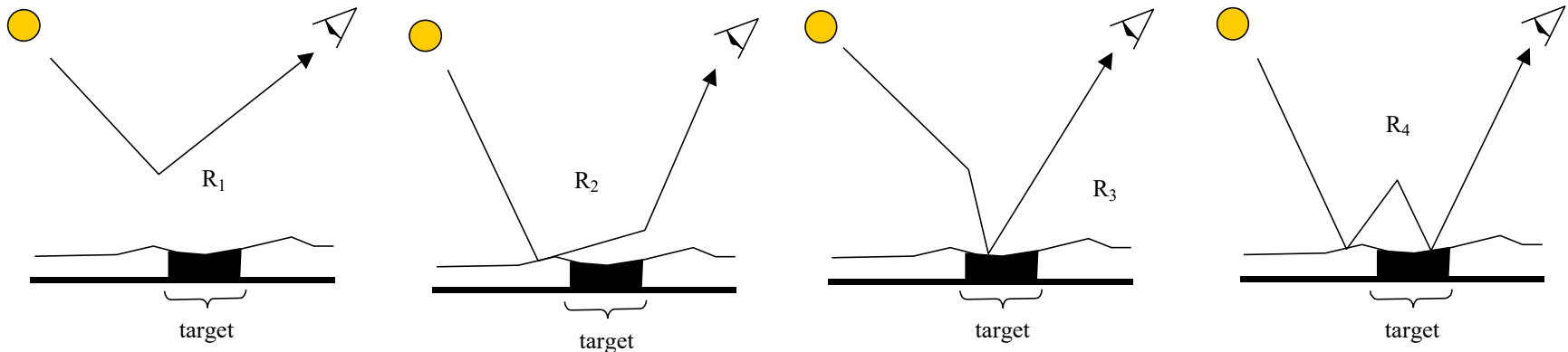
Attenuation



In-scattering



Atmospheric correction: simple



- Notice that target reflectance is a function of
 - Atmospheric irradiance (path radiance: R_1)
 - Reflectance outside target scattered into path (R_2)
 - Diffuse atmospheric irradiance (scattered onto target: R_3)
 - Multiple-scattered surface-atmosphere interactions (R_4)

Atmosphere correction

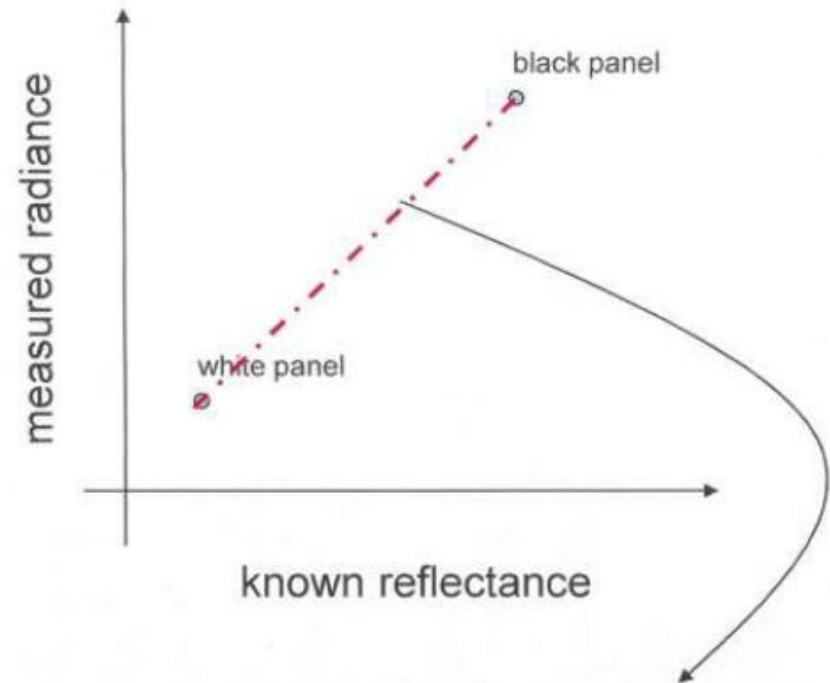
Most importantly: atmospheric influence



Importance
atmospheric
correction

Atmosphere correction

- Empirical (symples)
- Model based (FLAASH, MODTRAN, 6S)



Calibration function used to transform measured radiances (image) in to reflectance values!!!!

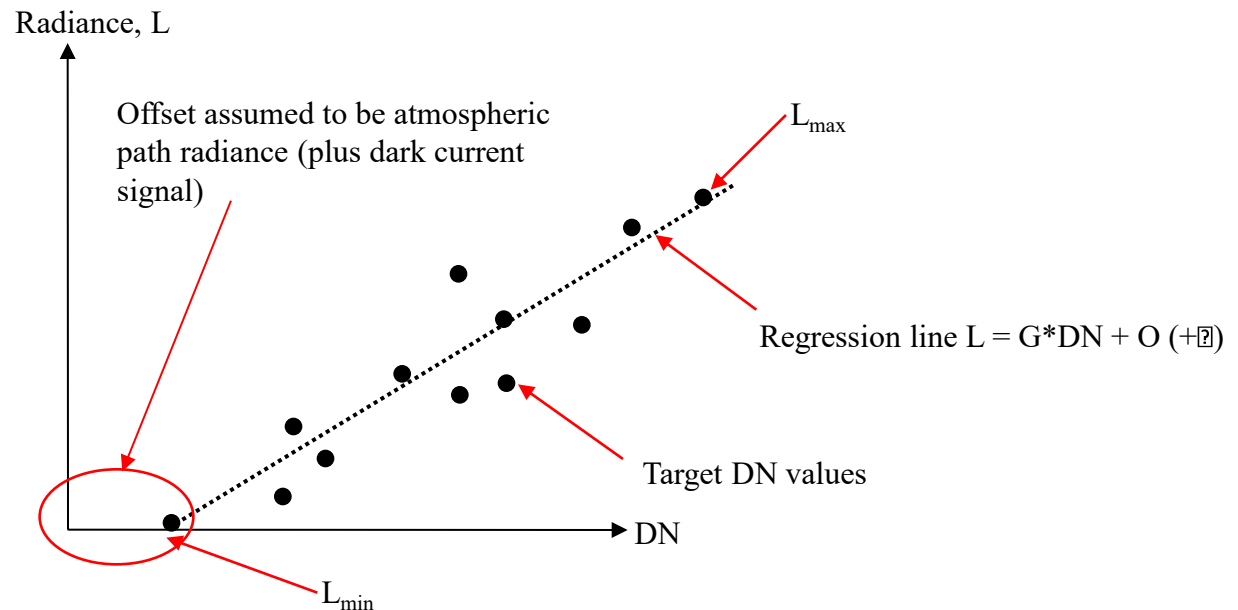
Atmospheric correction: simple

- So....need to remove impact of atmosphere on signal i.e. turn raw TOA DN into at-ground reflectance
- Simple methods?
 - Convert DN to apparent radiance L_{app} (sensor dynamic range)
 - Convert L_{app} to apparent reflectance (knowing response of sensor)
 - Convert to intrinsic surface property - at-ground reflectance in this case, by accounting for atmosphere

Atmospheric correction: simple

- Simple methods

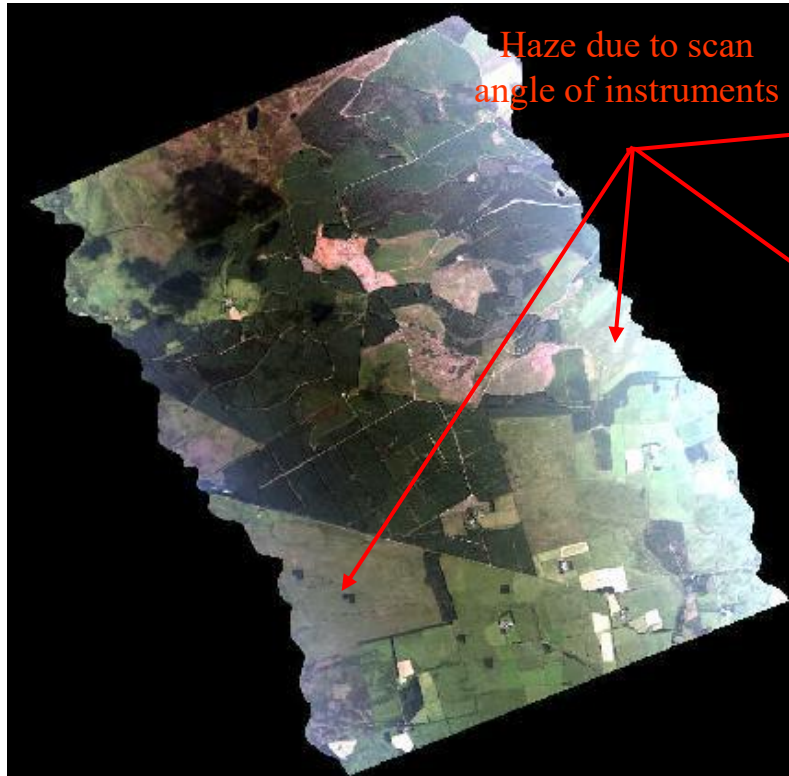
- e.g. empirical line correction (ELC) method
- Use target of “known”, low and high reflectance targets in one channel e.g. non-turbid water & desert, or dense dark vegetation & snow
- Assuming linear detector response, radiance, $L = \text{gain} * \text{DN} + \text{offset}$
- e.g. $L = \text{DN}(L_{\max} - L_{\min})/255 + L_{\min}$



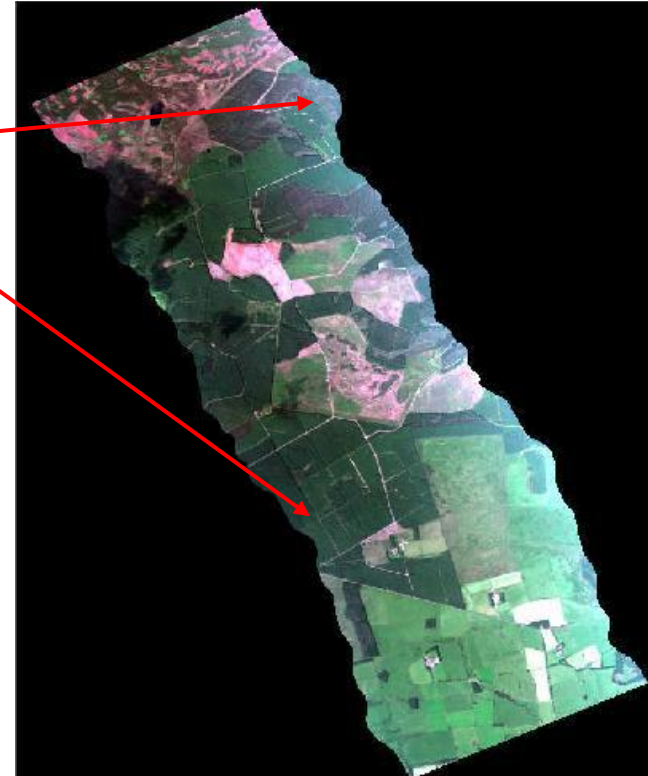
Atmospheric correction: simple

- Drawbacks
 - require assumptions of:
 - Lambertian surface (ignore angular effects)
 - Large, homogeneous area (ignore adjacency effects)
 - Stability (ignore temporal effects)
 - Also, per-band not per pixel so assumes
 - atmospheric effects invariant across image
 - illumination invariant across image
 - ok for narrow swath (e.g. airborne) but no good for wide swath

Example: airborne data



Airborne Thematic Mapper (ATM) data over Harwood Forest, Northumberland, UK, 13/7/2003



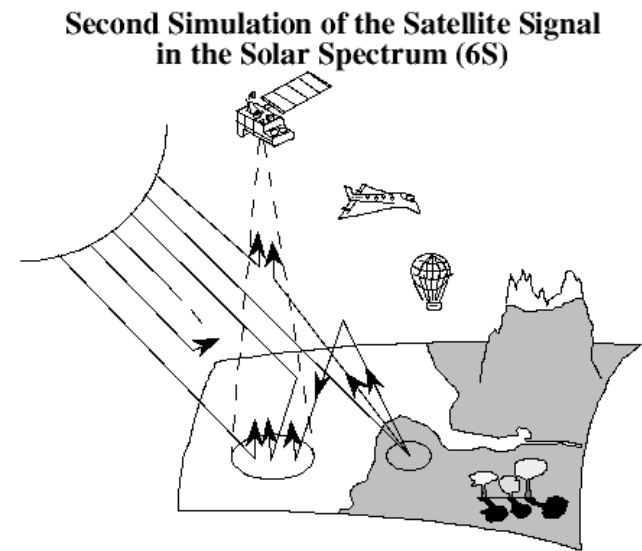
Compact Airborne Spectrographic Imager (CASI) data over Harwood Forest, Northumberland, UK, 13/7/2003

Atmospheric correction: complex

- Atmospheric radiative transfer modelling
 - use detailed scattering models of atmosphere including gas and aerosols
 - Second Simulation of Satellite Signal in Solar Spectrum (6s)
Vermote et al. (1997)
 - MODTRAN/LOWTRAN (Berk et al. 1998)
 - SMAC Rahman and Dedieu (1994)
 - FLAASH, ACORN, ATREM etc.

http://www-loa.univ-lille1.fr/Msixs/msixs_gb.html

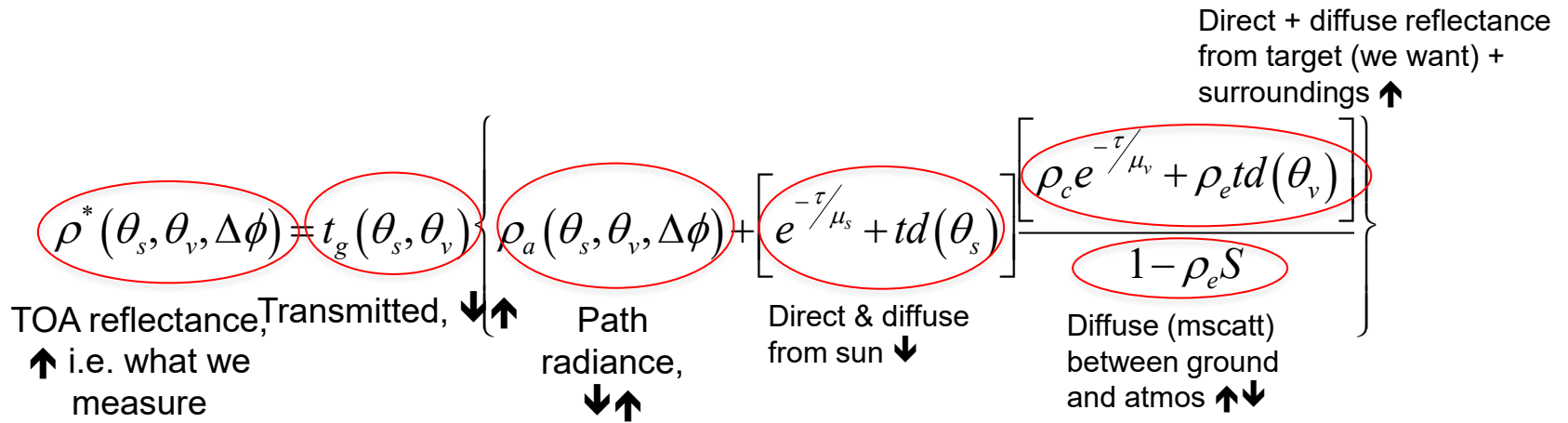
<http://geosci.uchicago.edu/~archer/cgimodels/radiation.html>



Vermote E. ^(1,*), Tanré D. ⁽²⁾, Deuzé J. L. ⁽²⁾,
Herman M. ⁽²⁾, and Morcrette J. J. ⁽³⁾

Atmospheric correction: complex

- 6S radiative transfer model: calculate upward and downward direct and diffuse fluxes



$\rho^*(\theta_s, \theta_v, \Delta\phi)$ = Top-of-atmosphere spectral reflectance, as a function of view and sun zenith $\theta_{s,v}$ and relative azimuth, $\Delta\phi$;

t_g = total gaseous transmission i.e. solar radiation to surface, then escaping on the way up;

ρ_a = atmospheric reflectance, function of molecular aerosols optical properties;

τ = atmos. optical depth ($e^{-\tau/\mu_s}$ and $e^{-\tau/\mu_v}$ = direct transmittance in sun & view directions, where μ_s, μ_v are $\cos(\theta_s)$ and $\cos(\theta_v)$ respectively;

$td(\theta_s), td(\theta_v)$ = diffuse transmittance in sun & view directions;

ρ_c = reflectance of target (what we want); ρ_e = reflectance of surrounding area;

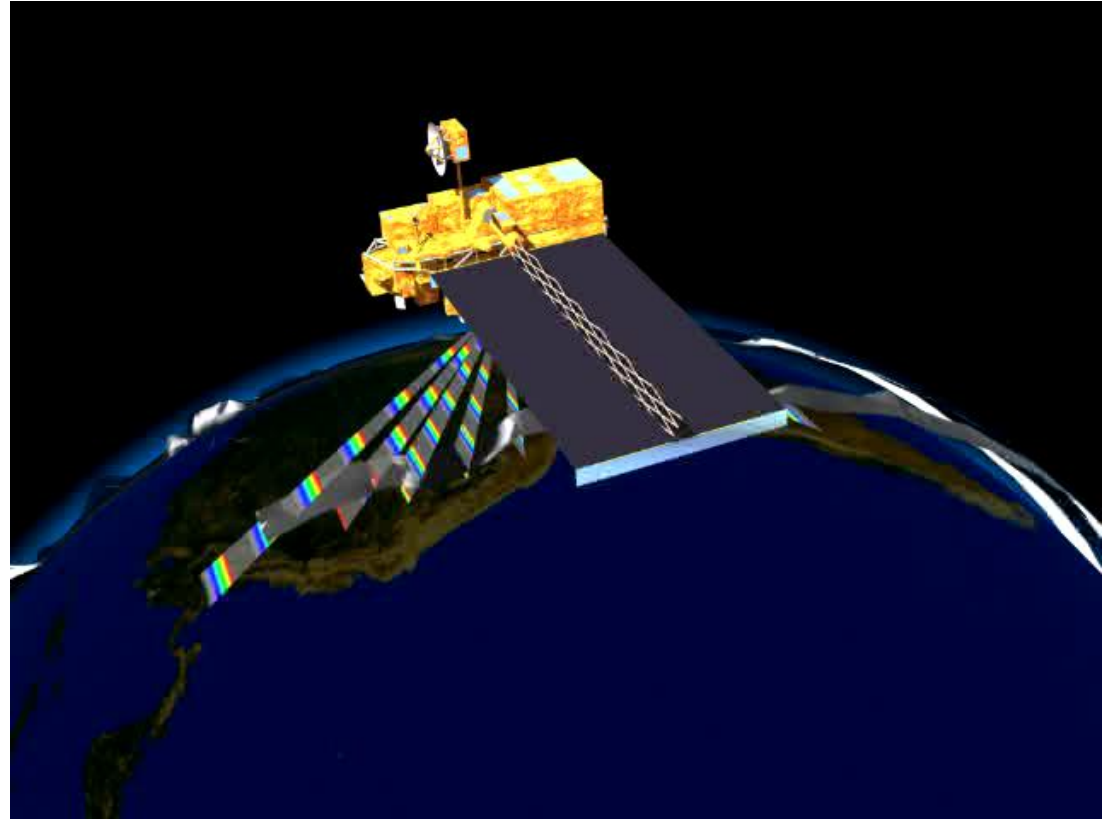
S = spherical (direct + diffuse) albedo of the atmosphere i.e. $1 - \rho_e S$ accounts for multiple scattering between ground (outside target) and atmosphere.....

Atmospheric correction: complex

- Radiative transfer models such as 6S require:
 - Geometrical conditions (view/illum. angles)
 - Atmospheric model for gaseous components (Rayleigh scattering)
 - H_2O , O_3 , aerosol optical depth, τ (opacity)
 - Aerosol model (type and concentration) (Mie scattering)
 - Dust, soot, salt etc.
 - Spectral condition
 - bands and bandwidths
 - Ground reflectance (type and spectral variation)
 - surface Bidirectional reflectance distribution function (BRDF) (default is to assume Lambertian....)
- If no info. use default values (Standard Atmosphere)

Atmospheric correction

- Can measure τ from ground and/or use multi-angle viewing to obtain different path lengths through atmos e.g. MISR, CHRIS
 - infer optical depth and path radiance AND aerosols
 - so use data themselves to infer atmos. scattering

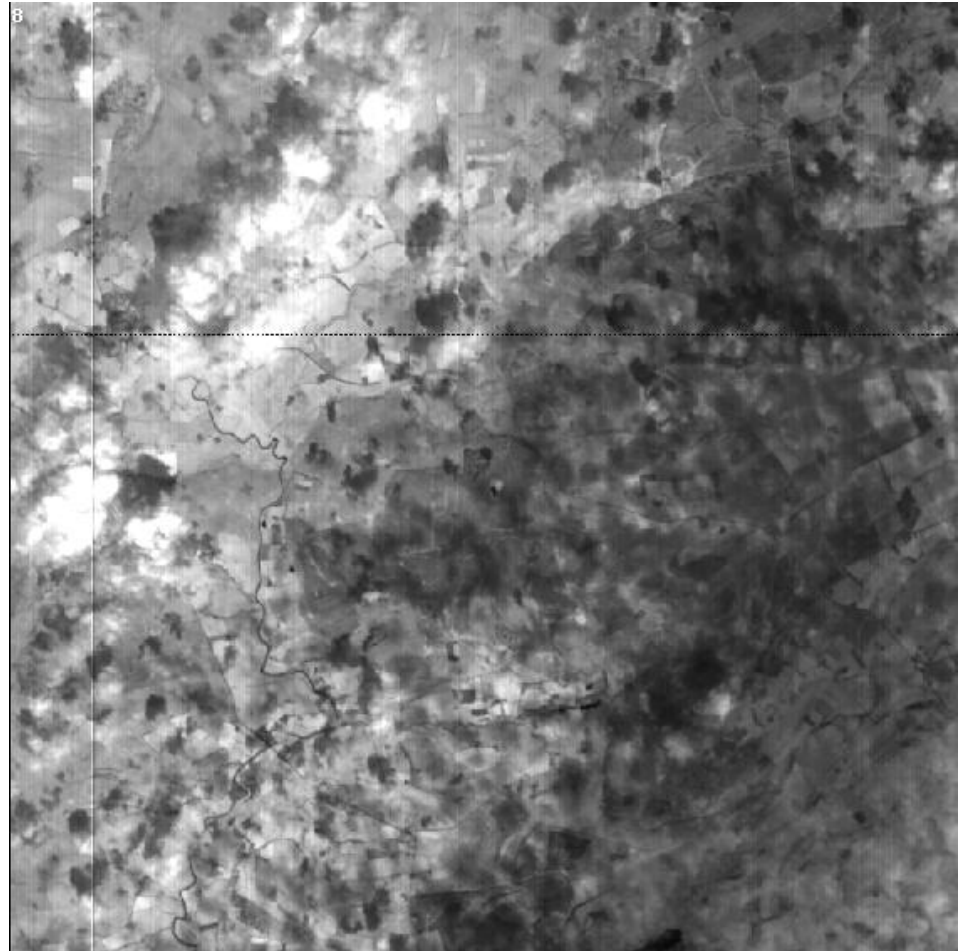


Atmospheric correction: summary

- Convert TOA radiance to at-ground reflectance
- VERY important to get right (can totally dominate signal)
- Simple methods
 - e.g. ELC but rough and ready and require many assumptions
- Complex methods
 - e.g. 6S but require much ancillary assumptions
 - BUT can use multi-angle measurements to correct
 - i.e. treat atmosphere as PART of surface parameter retrieval problem
 - different view angles give different PATH LENGTH

5.5 Radiometric correction

- Remove radiometric artifacts
 - dropped lines
 - detectors in CCD may have failed
 - fix by interpolating DNs

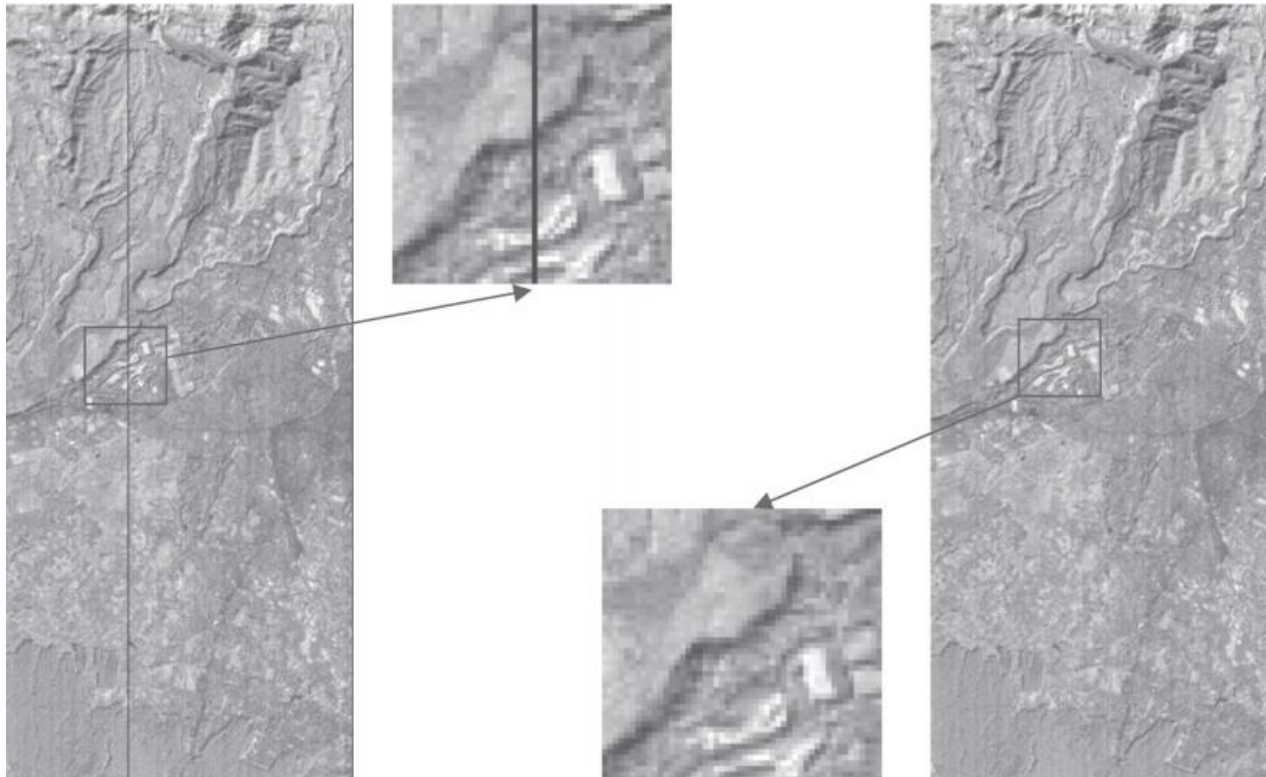


CHRIS-PROBA image over Harwood Forest,
Northumberland, UK, 9/5/2004

See <http://www.chris-proba.org.uk>

Radiometric correction

- Remove radiometric artifacts
 - striping
 - deterioration of detectors with time (& non-linearities)
 - Filter in Fourier domain to remove periodic striping



5.6 Image enhancement

- **Image enhancement process** consists of a collection of techniques that seek to improve the visual appearance of an image or to convert the image to a form better suited for analysis by a human or machine.
- **The principal objective** of image enhancement is to modify attributes of an image to make it more suitable for a given task and a specific observer.

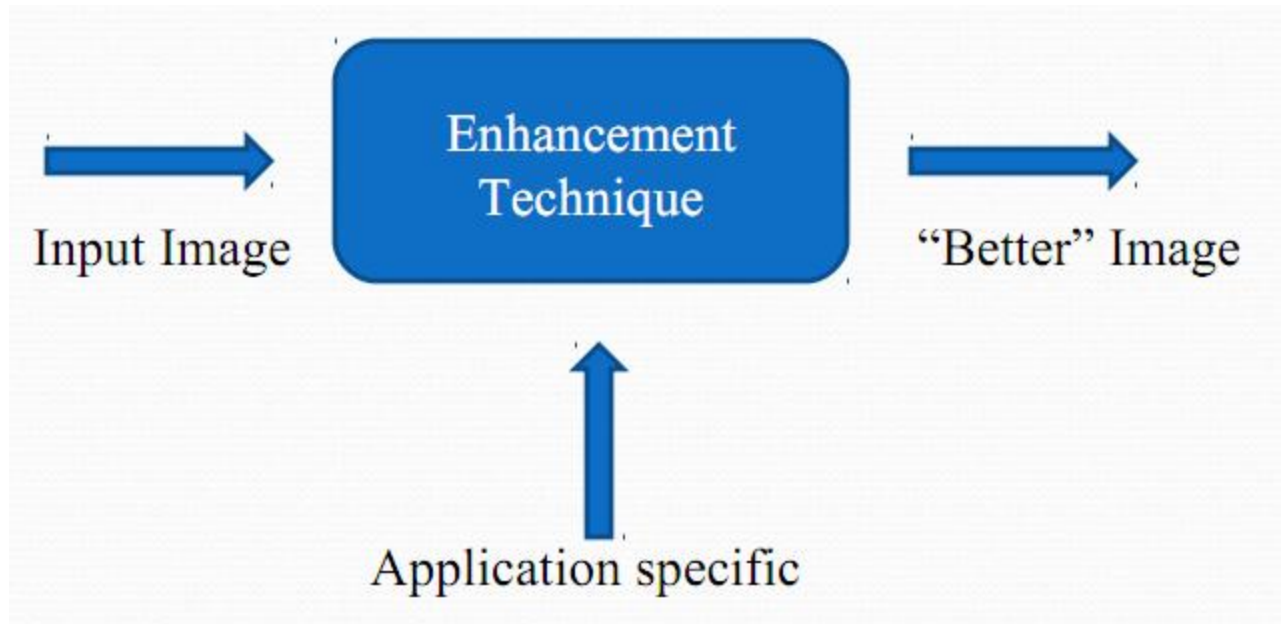
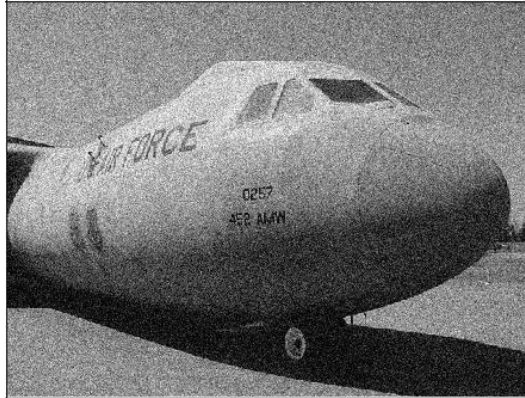


Image Enhancement

1. Noise removal



2. Contrast adjustment



Low contrast



Original contrast



High contrast

Contrast stretching

- Low-contrast images can result from poor illumination, lack of dynamic range in the image sensor, or even wrong setting of a lens aperture.
- Stretching is to increase the dynamic range of the gray levels in the image being processed.
- The general form is:

$$s = \frac{1}{1 + (m / r)^E}$$

where, r ; the input image values
 s ; are the output image values
 m ; is the threshold value and
 E ; the slope.

Noise reduction

- *Median Filtering* :

- The median filter is normally used to reduce noise in an image by preserving useful detail in the image.
- Each pixel in the image in turn and looks at its nearby neighbors to decide whether or not it is representative of its surroundings.
- Sorting all the pixel values from the surrounding neighborhood into numerical order and then replacing the pixel being considered with the middle pixel value

123	125	126	130	140
122	124	126	127	135
118	120	150	125	134
119	115	119	123	133
111	116	110	120	130

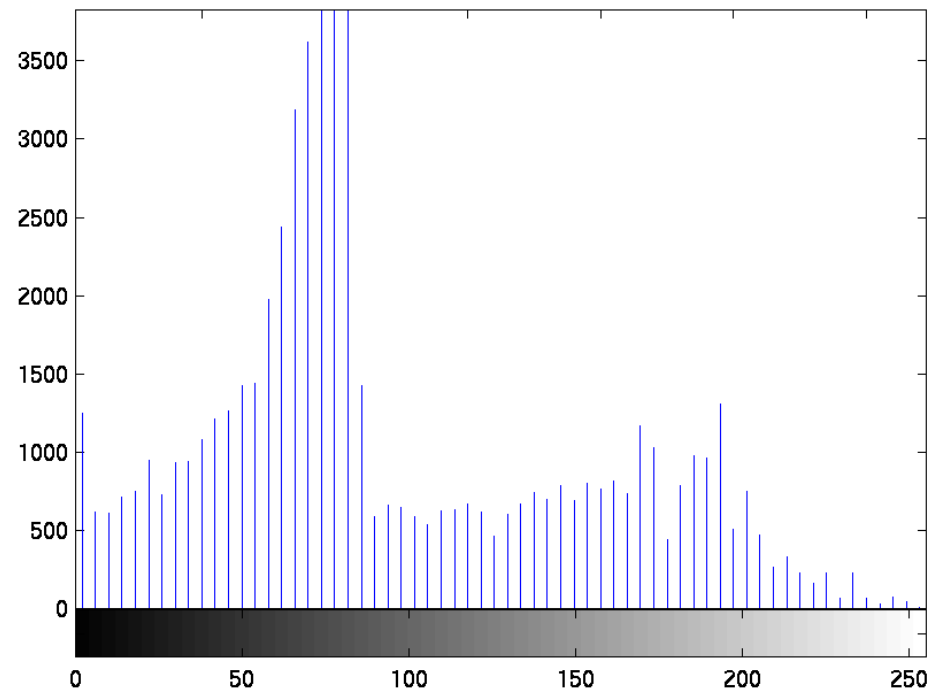
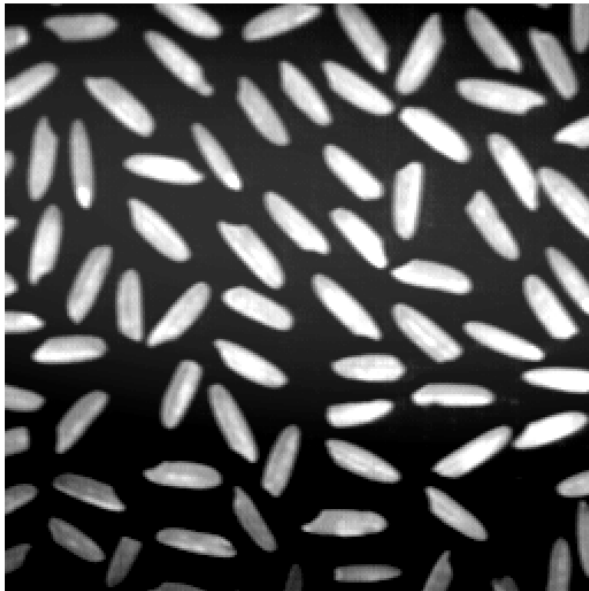
Neighbourhood values:

115, 119, 120, 123, 124,
125, 126, 127, 150

Median value: 124

Intensity Adjustment

Technique for mapping an image's intensity values to a new range.



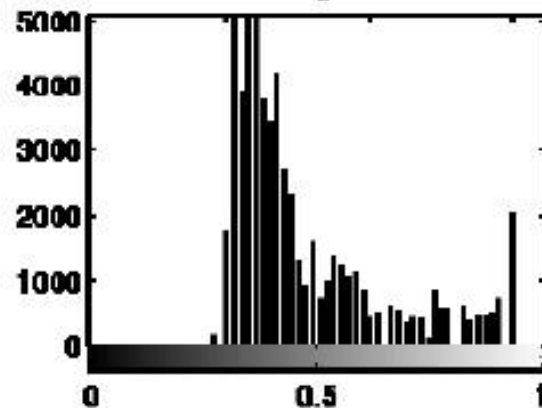
Histogram equalization

- generates a gray map which changes the histogram of an image and redistributing all pixels values to be as close as possible to a user – specified desired histogram

original



histogram



equalized image



equalized histogram

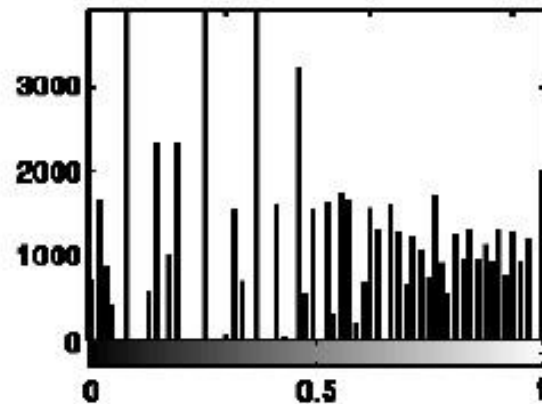


Image thresholding

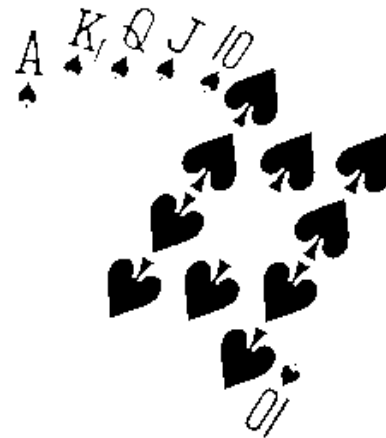
- Thresholding is the simplest segmentation method.
- The pixels are partitioned depending on their intensity value T .
- Global thresholding, using an appropriate threshold T :

$$g(x, y) = \begin{cases} 1, & \text{if } f(x, y) > T \\ 0, & \text{if } f(x, y) \leq T \end{cases}$$

- Imagine a poker playing robot that needs to visually interpret the cards in its hand:



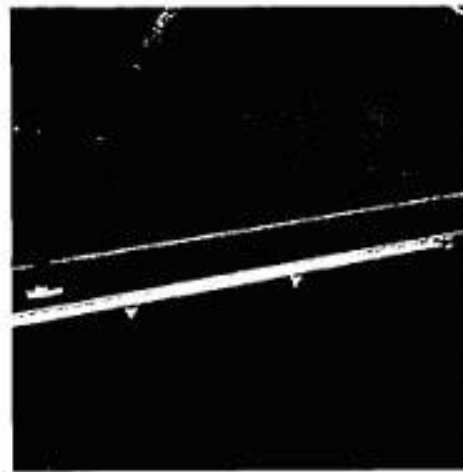
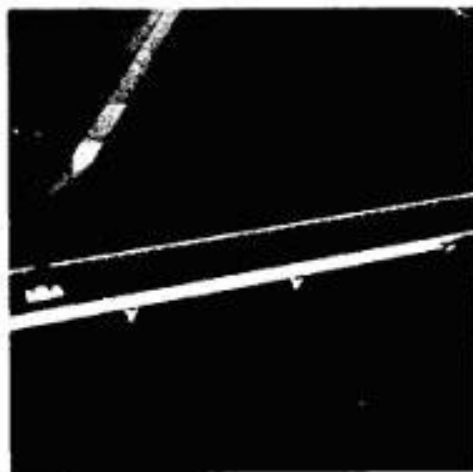
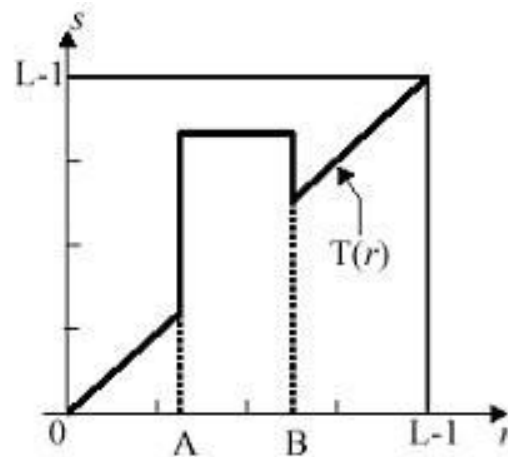
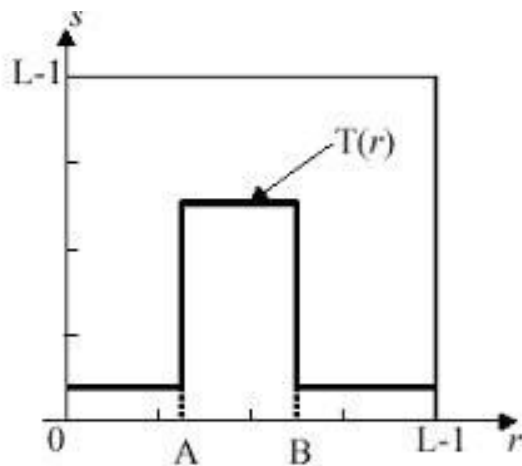
Original Image



Thresholded Image

Grey level slicing

- Grey level slicing is the spatial domain equivalent to band-pass filtering.
- A grey level slicing function can either emphasize a group of intensities and diminish all others or it can emphasize a group of grey levels and leave the rest alone.



Remarks

- Clearly define the pre-processing level of satellite data for appropriate applications.
- Applications related to spatial, temporal comparisons must use surface reflectance (Geometric, atmospheric, radiometric correction).

Thank you for your attention!